A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

Abstract. A countable discrete group is MF if it embeds in the unitary group of a matrix quotient $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C})/\bigoplus_n M_{d_n}(\mathbb{C})$; equivalently, if it admits finite-dimensional approximate unitary representations, asymptotically multiplicative in operator norm, that separate every nontrivial element from the identity. We construct an explicit finitely presented group E that is not MF, refuting the conjecture that every countable group is MF.

More generally, let Γ be a finitely presented property-(T) group with a proper injective endomorphism α. From (Γ, α) we build a finitely presented group carrying a central involution $w = u^2$, nontrivial because an explicit Clifford-group quotient sends w to -1, and we prove that $\Theta(w) = 1$ for every homomorphism Θ to $\mathcal{U}(\mathcal{Q})$. In every operator-norm asymptotic representation (U_n), one-sided conjugation preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm; the commutator u lies in that commutant, so $\|U_n(u) - 1\|_2 \to 0$. If $\Theta(w) \neq 1$, compressing to the (-1)-spectral projection of $\Theta(w)$ yields a contradiction: on that corner, taken with its own normalized trace, the image of w tends to -1 in operator norm while the image of u^2 tends to 1 in normalized Hilbert–Schmidt norm. Taking $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and $\alpha(v, A) = (2v, A)$ gives the eight-generator, forty-one-relator group E.

The group E is sofic, hence hyperlinear, so neither soficity nor hyperlinearity implies the MF property. Its reduced C^*-algebra $C^*_{\mathrm{r}}(E)$ is separable, exact and stably finite but not MF, and the canonical trace on $C^*_{\max}(E)$ is hyperlinear but not MF, which answers a question of Shulman negatively. The quotient $E/\langle w \rangle$ is also non-MF.

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1. Introduction

The *MF conjecture* (MF = matricial field) asks whether every countable discrete group admits finite-dimensional matrix models whose multiplicative errors tend to zero in operator norm and which asymptotically separate every nontrivial element from the identity [Sh]. The separation clause is part of the definition, since the constant identity maps are asymptotically multiplicative for every group. Throughout, MF for groups means the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE]; stronger conventions are discussed after Proposition 2.3. To our knowledge no counterexample was previously known [Sc, Sh]. We construct an explicit eight-generator, forty-one-relator group that is not MF. The eight generators display the affine action one generator at a time; a Tietze reduction, carried out in Section 7, brings the presentation down to six generators and thirty-two relators. Residually finite groups are MF because their finite quotients separate points; groups locally embeddable into finite groups (LEF) are MF [CDE], and quasidiagonality implies that amenable groups are MF [TWW]. Recent work establishes the MF property for many amalgamated free products [Sc, Sh, GKEMP].

The result.

Theorem A (an explicit finitely presented non-MF group)**.** *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 6.1, and put*

$$w = [tct^{-1},\, v_1(tct^{-1})v_1^{-1}].$$

The element w is central and satisfies $w^2 = 1$.

1. *$w \neq 1$ in E, so w is a central involution;*
2. *for every sequence (d_n) of positive integers and every homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q}((d_n)))$ into the unitary group of the corresponding matrix quotient, one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C^*_{\max}(E)$ nor $C^*_{\mathrm{r}}(E)$ is an MF C^*-algebra.

Outline of the proof. All approximation errors are measured in operator norm. Let Γ have property (T), let $\iota\colon \Gamma \to H$ be a homomorphism, and let $t, c \in H$ satisfy

$$t\,\iota(\Gamma)\,t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

Conjugation by unitaries turns an operator-norm asymptotic representation (U_n) of H into one on the Hilbert–Schmidt spaces $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$. These are the Hilbert spaces on which property (T) is applied, which is why a Hilbert–Schmidt norm governs a problem stated in operator norm; operator-norm control of the defects of (U_n) is what makes the conjugation actions asymptotically multiplicative, so that their classes multiply in a norm ultraproduct B_ω. Let h be the Hermitian average of a finite Kazhdan set of Γ acting through this representation. Property (T) gives a spectral gap for h, so the projection P onto the $\iota(\Gamma)$-invariant vectors is a spectral projection of h and belongs to B_ω. The inclusion above gives $P \le VPV^*$ for $V = [\operatorname{Ad} U_n(t)]_\omega$, and $V^*(VPV^*)$ exhibits the two projections as Murray–von Neumann equivalent. Since B_ω is a finite C^*-algebra, $P = VPV^*$. Hence conjugation by t preserves the asymptotic commutant of $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm (Theorem 3.4).

Put $\hat{c} = tct^{-1}$ and $u = [\,\hat{c}, \iota(a)\,]$ for $a \in \Gamma$. Since c commutes with $\iota(\Gamma)$, the sequence $(U_n(c))$ lies in that asymptotic commutant. Theorem 3.4 therefore gives $\|[U_n(\hat{c}), U_n(\iota(\gamma))]_-\|_2 \to 0$ for every $\gamma \in \Gamma$. Consequently, in every operator-norm asymptotic representation (U_n) of H,

$$\|U_n(u) - 1\|_2 \longrightarrow 0.$$

Suppose now that $w = u^2$ is central in H with $w^2 = 1$, and let Θ be a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w) \ne 1$. Then $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with $\Theta(H)$. After choosing unitary lifts, functional calculus gives projection lifts $q_n \in M_{d_n}(\mathbb{C})$ that are nonzero for infinitely many n. Compressing to these corners and giving each its own normalized trace gives an operator-norm asymptotic representation of H in which the image of w converges to -1 in operator norm. There the display above gives $\|U_n(u)^2 - 1\|_2 \to 0$, while $\|-1 - 1\|_2 = 2$. Hence $\Theta(w) = 1$, and if $w \ne 1$ then H is not MF (Theorem 4.2). Each corner has to carry its own trace: padding a model with a large identity block changes no operator norm and no multiplicative defect, while driving every normalized Hilbert–Schmidt distance to zero, so a 2-norm separation measured against d_n would prove nothing (Section S10).

Let Γ be a finitely presented Kazhdan group with a proper injective endomorphism α, let $a \in \Gamma \setminus \alpha(\Gamma)$, and adjoin t and c subject to $t\gamma t^{-1} = \alpha(\gamma)$, $c^2 = 1$ and $[c, \gamma] = 1$. In the direct limit of α the cosets $t\Gamma$ and $at\Gamma$ are distinct, since $t^{-1}at \in \Gamma$ would give $a \in \alpha(\Gamma)$. For a set X, the Clifford group $\mathrm{Cl}(X)$ is generated by a central involution ζ together with involutions c_x, $x \in X$, subject to $[c_x, c_y] = \zeta$ for $x \ne y$; every permutation of X acts on it by permuting the c_x and fixing ζ (Construction 8.1). Let the ambient group act in this way on $\mathrm{Cl}(X)$ for X the coset space, and send c to the generator c_x at the base coset. Then $\hat{c}$ and $a\hat{c}a^{-1}$ go to the generators at the two distinct cosets, so u goes to a product of two anticommuting involutions and $w = u^2$ to $\zeta \ne 1$ (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I)$$

gives the group E of Theorem A.

Property (T) is used only for the subgroup $\iota(\Gamma)$: the group E itself maps onto $\mathbb{Z}$ by sending t to 1 and every other generator to 0, and therefore does not have property (T). An exact finite-dimensional counterpart of the commutant argument, with equality of dimensions in place of finiteness of the ultraproduct, is Theorem 3.3.

The known implications among the finite approximation properties of countable groups are

$$\begin{array}{ccccc} \text{residually finite} & \Longrightarrow & \text{LEF} & \Longrightarrow & \text{MF} \\ & & \Downarrow & & \\ & & \text{sofic} & \Longrightarrow & \text{hyperlinear.} \end{array}$$

The converse of each arrow is open, as is MF $\Rightarrow$ hyperlinear (Question 2 of Section S10). The group E is sofic, hence hyperlinear, and not MF (Theorem D), and the canonical trace on $C^*_{\max}(E)$ is hyperlinear but not MF (Theorem E).

Relation to prior work.

MF algebras and traces. The MF problem for C^*-algebras predates the group question. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^*-algebra was known not to be MF [BK, CDE]. Their natural candidate was $C^*_{\mathrm{r}}(F_2)$, which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently implied the existence of abstract stably finite non-MF algebras [JNVWY, FGH]. To our knowledge no countable group was known whose reduced or maximal group C^*-algebra fails to be MF [Sh]. For maximal algebras the obstruction found here is not special to E: in any countable group with a property-(T) subgroup properly conjugated into itself, the Kazhdan projection of $C^*_{\max}$ is strictly dominated by its conjugate, so $C^*_{\max}$ contains a proper isometry and is not stably finite, hence not MF (Proposition S11.3).

There is a parallel hierarchy for traces. Brown developed tracial analogues of finite-dimensional approximation properties [Br], and MF traces were studied by Rainone–Schafhauser [RS]. We use Shulman's formulation [ShT], in which hyperlinear and MF traces differ by whether the algebraic defects are measured in normalized Hilbert–Schmidt norm or in operator norm. Several classes are known for which every hyperlinear trace is MF; for example, Shulman proves this for cones [ShC]. By contrast, the non-MF traces arising from the failure of the Connes embedding problem do not factor through tracial ultraproducts of matrix algebras and are not hyperlinear [JNVWY, Sc]. Theorem E separates the hyperlinear and MF trace classes.

Metric approximation problems. In his 2018 ICM address, Thom asked for which Schatten p-norms there exist non-approximable groups [Th, Question 3.11]. The case $p = 2$ was settled by De Chiffre, Glebsky, Lubotzky, and Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky and Oppenheim [LuO], and the case $p = 1$ by Bachner, Dogon, and Lubotzky [BDL]. Those results left three approximation problems open: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), or MF (unitaries in operator norm). The first was answered negatively in 2026: a nonsofic group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. The present paper answers the third negatively; as of August 2026 the hyperlinear problem remains open. The group-theoretic hypothesis is the one used there, a property-(T) subgroup Γ with $t\Gamma t^{-1} \subseteq \Gamma$. What it yields here is different: conjugation by t preserves the asymptotic commutant of Γ in normalized Hilbert–Schmidt norm, for every asymptotic representation whose defects are small in operator norm.

Related arguments. One-sided conjugation of a property-(T) subgroup, together with an element centralizing that subgroup, appears in the recent work on nonsofic groups [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the group acts by conjugation on the Hilbert–Schmidt space of each matrix block, and since the norm ultraproduct is a finite C^*-algebra, the projection onto the $\iota(\Gamma)$-fixed subspace equals its unitary conjugate (Theorem 3.4).

Slofstra's hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3]. He also considers the elements sent to the identity by every exact finite-dimensional representation and those sent to the identity by every approximate representation [Slo17, Definitions 2.5–2.7]; the two subgroups they form are the exact and the approximate analogues of the MF residual used below. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz's marked $\mathrm{BS}(2,3)$ example is the context for the cyclic comparison of Section S5 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in Section S3 [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5], and handle a finite normal subgroup by a character-isotypic corner [BDL, proof of Proposition 1.6]; Theorem S3.3 uses that corner together with one-sided conjugation of a property-(T) subgroup.

Bekka and Bekka–Valette study property (T) through the conjugation representation $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt space [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels' group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

MF for a group C^*-algebra implies MF for the underlying group, but the converse implication does not follow formally, and we know of no theorem establishing it or refuting it. For the algebra one asks for an embedding $C^*_{\mathrm{r}}(E) \hookrightarrow \mathcal{Q}$ of C^*-algebras; for the group one asks only for an injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$, which need not extend to $C^*_{\mathrm{r}}(E)$. Failure of MF for a group C^*-algebra therefore leaves open whether the group itself embeds. For E, Theorem A rules out both.

2. Matrix quotients

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. For group elements $[g,h]=ghg^{-1}h^{-1}$; for operators $[x,y]_-=xy-yx$. Throughout, $*$-homomorphisms are not assumed unital unless explicitly stated; in particular, an MF embedding may be nonunital. For a unital C^*-algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r, the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2=\mathrm{tr}_r(x^*x)^{1/2}$. For a nonzero projection $q\in M_r(\mathbb{C})$ we equip the corner $qM_r(\mathbb{C})q$ with its own normalized trace. When the trace must be specified, we write $\|x\|_{2,\mathrm{tr}_r}$. The unnormalized trace is $\mathrm{Tr}=r\cdot\mathrm{tr}_r$ on $M_r(\mathbb{C})$. We repeatedly use the inequalities $\|x\|_2\le\|x\|$; $|\mathrm{tr}_r(x)|\le\|x\|$; and $\|uxv\|_2=\|x\|_2$ for unitaries u,v.

Given a sequence $(d_n)_{n\in\mathbb{N}}$ of positive integers, write

$$\mathcal{Q}((d_n)_{n\in\mathbb{N}}) \;=\; \prod_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}) \Big/ \bigoplus_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}), \tag{1}$$

where the product is the bounded C^*-product and $\bigoplus_n$ is the c_0-sum, the ideal of sequences with $\|x_n\|\to 0$; equivalently, $\mathcal{Q}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$. Write $q\colon \prod_n M_{d_n}(\mathbb{C})\to\mathcal{Q}$ for the quotient map.

Following Blackadar–Kirchberg [BK], a separable C^*-algebra is *MF* if it embeds into an algebra of the form (1).

Convention on the dimension sequence. When the dimension sequence is not displayed, $\mathcal{Q}$ denotes $\mathcal{Q}((d_n)_{n\in\mathbb{N}})$ for a sequence fixed by the surrounding discussion. If no sequence has been fixed, a map into $\mathcal{Q}$ or $\mathcal{U}(\mathcal{Q})$ is understood to allow the sequence (d_n) to vary; likewise, a quantification over all such maps also ranges over all dimension sequences. Proposition 2.3 shows that nothing changes if the dimensions are required to be strictly increasing.

Three quotients of $\prod_n M_{d_n}(\mathbb{C})$ appear below, differing in which sequences they annihilate:

- $\mathcal{Q}$, where $\|x_n\|\to 0$: the norm matrix corona (1), and the target of an MF embedding;
- B_ω, where $\lim_\omega\|x_n\|=0$: the norm ultraproduct of Section 3, a finite C^*-algebra, in which the Kazhdan projection lives;
- the tracial ultraproduct, where $\lim_\omega\|x_n\|_2=0$: the target for hyperlinearity.

The first two see the operator norm and the third does not, which is why the argument below fails in the third (Section S10).

The following lifting lemmas are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 2.1 (lifting)**.** *Every unitary $u\in\mathcal{Q}$ lifts to a sequence of unitaries.*

Proof. Lift u to a bounded sequence (x_n); unitarity of $q((x_n))$ gives $\|x_n^*x_n-1\|\to 0$.Once $\|x_n^*x_n-1\|\le\frac{1}{2}$, the matrix x_n is invertible and the polar correction $u_n = x_n(x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \le 2\|x_n\|\,\|x_n^* x_n - 1\| \to 0$ by continuous functional calculus;set $u_n = 1$ at the finitely many remaining indices. $\square$

Write

$$\mathcal{N} = \{(u_n) \in \textstyle\prod_n \mathcal{U}(d_n) \ : \ \|u_n - 1\| \longrightarrow 0\}, \tag{2}$$

a normal subgroup of $\prod_n \mathcal{U}(d_n)$.

Lemma 2.2 (unitaries in the quotient)**.** *For every sequence (d_n), with $\mathcal{N}$ as in (2), the map*

$$\kappa_{(d_n)} \colon \prod_n \mathcal{U}(d_n)/\mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \qquad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. Thus a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ may be represented by unitary lifts $U_n(g)$ whose multiplicative defects tend to 0 in operator norm, with two lift sequences identified when their quotient tends to 1.

Proof. The kernel of $(u_n) \mapsto q((u_n))$ is exactly $\mathcal{N}$, so the induced map is injective.Surjectivity is Lemma 2.1. $\square$

An *operator-norm asymptotic representation* of H is a sequence of maps $U_n \colon H \to \mathcal{U}(d_n)$ such that $\|U_n(g)U_n(h) - U_n(gh)\| \to 0$ for all $g, h \in H$. Taking $g = h = 1$ gives $\|U_n(1) - 1\| \to 0$ automatically, since $U_n(1)$ is unitary. By Lemma 2.2, choosing unitary lifts $U_n(g)$ of $\Theta(g)$ turns a homomorphism $\Theta \colon H \to \mathcal{U}(\mathcal{Q})$ into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ for some sequence of dimensions.

For a finite set $F \subseteq H$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$, an *approximate unitary representation* is a map $V \colon H \to \mathcal{U}(r)$, for some $r \ge 1$, such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\le \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\ge \delta \quad (g \ne h \in F). \end{aligned}$$

Proposition 2.3 (Equivalent formulations of the MF property)**.** *For every countable group, the group-MF definition above — an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ — is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

The proof of Proposition 2.3 is given in Section S1. Two stronger conventions also occur in the literature: in [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring element, and the purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

3. One-sided conjugation in matrix models

An inclusion whose two sides have the same size is an equality. For a homomorphism π to a finite group, $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$, since the left side is a conjugate of the right side contained in it and the two are finite of the same cardinality. In finite dimensions, the commutant C of the base satisfies $C \subseteq \pi(t)C\pi(t)^{-1}$; equality follows because the two spaces have the same finite dimension. In the norm-ultraproduct argument, the fixed-space projection satisfies $P \leq VPV^*$ by $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$; the two projections are Murray–von Neumann equivalent, and finiteness of the ultraproduct gives $P = VPV^*$.

Lemma 3.1 (comparison in a finite algebra)**.** *Let A be a finite unital C^*-algebra and let $p, q \in A$ be projections with $p \leq q$ and $p \sim q$. Then $p = q$.*

Proof. Let $r \in A$ satisfy $r^*r = q$ and $rr^* = p$. From $r^*r = q$ we get $r = rq$. Since $p \leq q$ we have $qp = p$, so

$$pr = rr^*r = rq = r, \qquad qr = q(pr) = (qp)r = pr = r,$$

and $r = qr$ as well. Hence the cross terms in $\sigma^*\sigma$ vanish for $\sigma = r + (1-q)$, and $\sigma^*\sigma = q + (1-q) = 1$, so σ is an isometry. Since A is finite, σ is unitary, and $\sigma\sigma^* = p + (1-q) = 1$ gives $p = q$. $\square$

Lemma 3.2 (finiteness of a norm quotient of matrix blocks)**.** *Let (K_n) be finite-dimensional Hilbert spaces and let*

$$B_{\mathrm{c}} = \prod_n B(K_n) \Big/ \bigoplus_n B(K_n)$$

be the quotient of the bounded family algebra by the c_0-sum. Then B_{c} is a finite C^*-algebra: every $\sigma \in B_{\mathrm{c}}$ with $\sigma^*\sigma = 1$ satisfies $\sigma\sigma^* = 1$.

Proof. Lift σ to a bounded family (σ_n). Then $\|\sigma_n^*\sigma_n - 1\| \to 0$, so for all large n the Gram defect is at most $\frac{1}{2}$ and $\sigma_n^*\sigma_n$ is invertible; each σ_n acts on a finite-dimensional space, so polar correction replaces it by a unitary $w_n = \sigma_n(\sigma_n^*\sigma_n)^{-1/2}$ with $\|\sigma_n - w_n\| \to 0$. Hence σ is the class of (w_n), and $w_n w_n^* = 1$ at every such n gives $\sigma\sigma^* = 1$. $\square$

Let ω be a free ultrafilter on $\mathbb{N}$, let K_ω be the Hilbert-space ultraproduct of the K_n, and let $B_\omega = \prod_\omega B(K_n)$ be the norm ultraproduct, acting on K_ω by $[A_n]_\omega[\xi_n]_\omega = [A_n\xi_n]_\omega$. The same polar-correction argument shows that B_ω is finite, with ordinary eventual convergence replaced by convergence along ω. Its action on K_ω is faithful: if $A = [A_n]_\omega \neq 0$ then $\lim_\omega \|A_n\| = \delta > 0$, and unit vectors ξ_n with $\|A_n\xi_n\| \geq \|A_n\| - 1/n$ give $\|A[\xi_n]_\omega\| = \delta$. Hence for projections $P, Q \in B_\omega$ the inclusion $\operatorname{ran} P \subseteq \operatorname{ran} Q$ makes QP and P act identically on K_ω, so $QP = P$, that is, $P \leq Q$; the converse is immediate.

The exact case.

Theorem 3.3 (the finite-dimensional case)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, $\Gamma \leq H$ a subgroup, $t, c \in H$, and $a \in \Gamma$. Assume that $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi\colon H \to \mathrm{GL}(V)$ satisfies*

$$\pi\big([\, tct^{-1},\ a(tct^{-1})a^{-1}]\big) \ =\ 1.$$

Proof. Let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$, and write $\Phi(x) = \pi(t)\,x\,\pi(t)^{-1}$ for conjugation by $\pi(t)$, a linear automorphism of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and put $y = \pi(t)^{-1}x\,\pi(t)$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x, and therefore

$$\pi(\gamma)\,y = \pi(t)^{-1}\,\pi(t\gamma t^{-1})\,x\,\pi(t) = \pi(t)^{-1}\,x\,\pi(t\gamma t^{-1})\,\pi(t) = y\,\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$C \ =\ \Phi(C) \ =\ \pi(t)\,C\,\pi(t)^{-1}. \tag{3}$$

Now $\pi(c) \in C$ because c centralizes Γ, so $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$ by (3). Since $a \in \Gamma$, the operator $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi\big(a(tct^{-1})a^{-1}\big) = \pi(tct^{-1})$ and

$$\pi\big([\, tct^{-1},\ a(tct^{-1})a^{-1}]\big) = \big[\,\pi(tct^{-1}),\ \pi(tct^{-1})\big] = 1. \qquad \square$$

Every finite-dimensional linear representation of E over any field maps w to the identity; consequently every homomorphism from E to a finite group does as well (Corollary S4.2).

In an MF model, multiplicativity is only asymptotic, so the coordinate fixed spaces need not be invariant. Because the multiplicative defects tend to 0 in operator norm, the adjoint actions on the Hilbert–Schmidt spaces are asymptotically multiplicative, and the Kazhdan projection they carry lies in a finite norm ultraproduct, where $P \leq VPV^*$ and $P \sim VPV^*$ imply $P = VPV^*$. The property (T) hypothesis is essential: the group E_{BS} of Section S5 has cyclic base, its distinguished word is trivial in every finite-dimensional representation over every field, and some homomorphism $E_{\mathrm{BS}} \to \mathcal{U}(\mathcal{Q})$ maps that word to a nontrivial element.

Theorem 3.4 (one-sided conjugation preserves asymptotic commutants)**.** *Let Γ and H be groups, let Γ have property (T), let $\iota\colon \Gamma \to H$ be a homomorphism, let $s \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and asymptotically centralizes $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm,

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma))x_n\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma),$$

then the conjugated sequence does as well:

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]_-\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma).$$

Proof. Suppose the conclusion fails: there are $\gamma_0 \in \Gamma$, $\delta > 0$, and an infinite set $I \subseteq \mathbb{N}$ with $\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma_0))]_-\|_2 \geq \delta$ for $n \in I$. Fix a free ultrafilter ω on $\mathbb{N}$ with $I \in \omega$.

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $K_n = L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and let each group element act on it by conjugation, $\operatorname{Ad} U_n(g)\,\xi = U_n(g)\,\xi\,U_n(g)^*$. The operator-norm almost multiplicativity of U_n implies that $g \mapsto \operatorname{Ad} U_n(g)$ is an operator-norm asymptotic representation on these Hilbert spaces.

The ultraproduct. Let K_ω and $B_\omega = \prod_\omega B(K_n)$ be as in Section 3; write $[\,\cdot\,]_\omega$ for classes in either. The action of B_ω on K_ω is faithful, and B_ω is finite: every isometry in it is unitary. For projections in B_ω, inclusion of ranges in K_ω is equivalent to the projection order. Asymptotic multiplicativity implies that

$$\pi\colon H \longrightarrow \mathcal{U}(B_\omega), \qquad \pi(g) = [\operatorname{Ad} U_n(g)]_\omega,$$

a homomorphism.

The projection onto the fixed subspace. Choose a finite symmetric Kazhdan set $S \subseteq \Gamma$ containing 1, with Kazhdan constant $\kappa \in (0, 1]$. Let h be the Hermitian part of $\frac{1}{|S|}\sum_{s' \in S} \pi(\iota(s'))$ and let $\mathrm{Fix} \subseteq K_\omega$ be the subspace of vectors fixed by every $\pi(\iota(\gamma))$. Both Fix and its orthocomplement are invariant under each unitary $\pi(\iota(s'))$, hence under h. On Fix, $h = 1$. The restriction of $\pi \circ \iota$ to $\mathrm{Fix}^\perp$ has no nonzero invariant vector, so a unit vector $\xi \in \mathrm{Fix}^\perp$ satisfies $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$ for some $s' \in S$. For this s' $\operatorname{Re}\langle \pi(\iota(s'))\xi, \xi\rangle \leq 1 - \kappa^2/2$, while every other term is at most 1. Averaging gives $\langle h\xi, \xi\rangle \leq 1 - \kappa^2/(2|S|)$. Consequently

$$\mathrm{sp}(h) \subseteq \left[-1,\, 1 - \tfrac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection P of h at the isolated point 1 belongs to B_ω by continuous functional calculus and projects onto Fix.

The two projections agree. Let $V = \pi(s)$ and $Q = VPV^*$. Since $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$, every vector of Fix is fixed by each $\pi(s\iota(\gamma)s^{-1})$; and η is fixed by all $\pi(s\iota(\gamma)s^{-1})$ exactly when $V^*\eta \in \mathrm{Fix}$. Hence $\operatorname{ran} P = \mathrm{Fix} \subseteq V\,\mathrm{Fix} = \operatorname{ran} Q$, so $P \leq Q$. The element $r = V^*Q$ satisfies $r^*r = Q$ and $rr^* = V^*QV = P$, so $P \sim Q$. Since B_ω is finite, Lemma 3.1 gives $Q = P$.

Conclusion. Let $\xi = [x_n]_\omega \in K_\omega$; the uniform operator-norm bound makes this class well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each $\pi(\iota(\gamma))$ fixes

ξ. Thus $\xi \in \mathrm{Fix}$. Since $Q = P$, we obtain $V\xi \in V\,\mathrm{Fix} = \mathrm{ran}\,Q = \mathrm{ran}\,P = \mathrm{Fix}$. The matrix $U_n(s)x_nU_n(s)^*$ is the vector $\mathrm{Ad}\,U_n(s)\,x_n$ of K_n, so, again by unitary invariance,

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]_-\|_2 = \|(\mathrm{Ad}\,U_n(\iota(\gamma)) - 1)\ \mathrm{Ad}\,U_n(s)\,x_n\| \longrightarrow 0$$

along ω, for every $\gamma \in \Gamma$. For $\gamma = \gamma_0$ this contradicts $I \in \omega$. □

4. The central involution obstruction

Let w be a central involution of H and let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, and $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection on whose corner $q\mathcal{Q}q$ the compression of $\Theta(w)$ is $-q$ exactly. The argument below, and the generalizations in Section S3, compress a matrix model of H to the corners cut out by coordinate lifts of such a projection.

Lemma 4.1 (compression to an asymptotically central corner)**.** *Let H be a group, let $V_{g,n} \in \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H, and let $q_n \in M_{d_n}(\mathbb{C})$ be nonzero projections with*

$$\|[q_n, V_{g,n}]_-\| \longrightarrow 0 \qquad (g \in H).$$

Write $r_n = \mathrm{rank}\,q_n \geq 1$ and identify $q_nM_{d_n}(\mathbb{C})q_n \cong M_{r_n}(\mathbb{C})$. Then there are unitaries $W_{g,n} \in \mathcal{U}(r_n)$ with

$$\|W_{g,n} - q_nV_{g,n}q_n\| \longrightarrow 0 \qquad (g \in H),$$

and $(W_{g,n})$ is an operator-norm asymptotic representation of H on the corners.

Proof. Fix g, abbreviate $e = q_n$, $a = V_{g,n}$ and $\delta = \|[e,a]_-\|$, and note that the unit of the corner is e. From $eae - ae = [e,a]_-e$ and $ea^* - a^*e = -([e,a]_-)^*$ we get $\|eae - ae\| \leq \delta$ and $\|ea^* - a^*e\| \leq \delta$, so

$$\|(eae)^*(eae) - e\| = \|ea^*eae - a^*ae\| \leq 2\delta,$$

since $a^*a = 1$. Once $2\delta \leq \frac{1}{2}$ the corner element eae is invertible in $M_{r_n}(\mathbb{C})$, and its polar correction $W_{g,n} = eae\left((eae)^*(eae)\right)^{-1/2}$ is unitary with $\|W_{g,n} - eae\| \leq 4\delta$, by the estimate in the proof of Lemma 2.1; set $W_{g,n} = 1$ at the finitely many remaining indices. For $g, h \in H$ write $b = V_{h,n}$ and $\varepsilon = \|ab - V_{gh,n}\|$. Then

$$\|(eae)(ebe) - eV_{gh,n}e\| \leq \|ea(eb - be)e\| + \|e(ab - V_{gh,n})e\| \leq \|[e,b]_-\| + \varepsilon,$$

which tends to 0. Adding the three polar corrections gives

$$\|W_{g,n}W_{h,n} - W_{gh,n}\| \longrightarrow 0.$$ □

Each corner block is given its own normalized trace tr_{r_n}, not the restriction of tr_{d_n}, so that $\|1 - (-1)\|_{2,\mathrm{tr}_{r_n}} = 2$ whatever the ratio r_n/d_n; Section S10 shows why this renormalization is necessary.

Theorem 4.2 (central involution obstruction)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

Put $\hat{c} = tct^{-1}$ and $u = [\hat{c}, \iota(a)]$ for some $a \in \Gamma$. If $w := u^2$ is a nontrivial central involution of H, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps w to the identity, and H is not MF.

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and suppose $\Theta(w) \neq 1$.

The (-1)*-spectral corner.* Choose coordinate unitary lifts $(V_{g,n})$ of Θ (Lemma 2.2) and put $p_n = \frac{1}{2}(1 + V_{w,n})$, the averaging operator of the two-element subgroup $F = \{1, w\}$. Because the lifts are asymptotically multiplicative and $w^2 = 1$, the Hermitian part of p_n is asymptotically idempotent. Its spectrum therefore concentrates near $\{0, 1\}$, and continuous functional calculus at $\frac{1}{2}$ gives projections at operator-norm distance $o(1)$ from p_n. Since w is central, these projections asymptotically commute with every lift of $\Theta(H)$. Let q_n be their complements. If q_n vanished for all large n the averages would converge to 1 and $\Theta(w)$ would be 1; so $q_n \neq 0$ along infinitely many coordinates, which we retain and relabel by $\mathbb{N}$. Lemma 4.1 now gives an operator-norm asymptotic representation $(W_{g,n})$ of H on the corners $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$. On those corners $q_n V_{w,n} q_n = -q_n + o(1)$, because q_n is the complement of a projection within $o(1)$ of $\frac{1}{2}(1 + V_{w,n})$, so

$$\|W_{w,n} + 1\| \longrightarrow 0.$$

Equip each corner block with its own normalized trace tr_{r_n}; all estimates below are relative to r_n and independent of the ratio r_n/d_n.

The commutant on the corner. Since c centralizes $\iota(\Gamma)$, the sequence $(W_{c,n})$ centralizes $(W_{\iota(\gamma),n})$ asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. By Theorem 3.4 applied to $x_n = W_{c,n}$, the conjugated sequence $(W_{\hat{c},n})$ lies in the same asymptotic commutant. Therefore $\|W_{u,n} - 1\|_2 \to 0$, and squaring gives $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$.

The contradiction. The relation $w = u^2$ gives $\left\|W_{u,n}^2 - W_{w,n}\right\| \to 0$, while $W_{w,n} \to -1$ in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence $\left\|W_{u,n}^2 + 1\right\|_2 \to 0$ and $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$. The triangle inequality would then give $\|2 \cdot 1\|_2 \to 0$, contradicting $\|2 \cdot 1\|_2 = 2$. Thus $\Theta(w) = 1$. Since $w \neq 1$, every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ has nontrivial kernel, so H is not MF. □

Here $\{1, w\}$ can be replaced by any finite normal subgroup of the normal closure of the commutators $[tct^{-1}, \iota(\gamma)]$, $\gamma \in \Gamma$ (Theorem S3.3), and, with a Kazhdan spectral projection in place of the two-element average, by any normal property-(T) subgroup of that closure, which may be infinite, noncentral and torsion-free (Theorem S3.7).

5. The Kazhdan–Clifford construction

Theorem 5.1 (Kazhdan–Clifford construction)**.** *Let Γ be a finitely presented group with property* (T)*, let $\alpha\colon \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \qquad c^2 = 1, \qquad [c,\gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$\hat{c} = tct^{-1}, \qquad u = [\hat{c}, a], \qquad w = u^2 = [\hat{c}, a\hat{c}a^{-1}],$$

and impose centrality of w. More explicitly, for a finite generating set S_Γ it is enough to impose the conjugation and commutation relations for $s \in S_\Gamma$, together with

$$[w,s] = 1 \quad (s \in S_\Gamma), \qquad [w,t] = [w,c] = 1.$$

The group $E(\Gamma,\alpha,a)$ is finitely presented, the canonical map $\Gamma \to E(\Gamma,\alpha,a)$ is injective, and w is a nontrivial central involution. Every homomorphism $E(\Gamma,\alpha,a) \to \mathcal{U}(\mathcal{Q})$ maps w to 1, so $E(\Gamma,\alpha,a)$ is not MF.

Proof. Finite presentability follows from the relative presentation. Start with the free product of Γ and the free group on t, c, and quotient by the finitely many displayed relations for a finite generating set of Γ.

The relation $w^2 = 1$ is redundant. Put $b = a\hat{c}a^{-1}$. Both $\hat{c}$ and b are involutions, and $w = [\hat{c}, b] = (\hat{c}b)^2$. A direct computation gives

$$dwd^{-1} = w^{-1}. \tag{4}$$

Centrality also gives $dwd^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma)\rangle$$

concretely. Let Γ_∞ be the direct limit of the system $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$, whose nth copy of Γ we call its *nth level*; injectivity of α passes to the limit, so the level-zero map embeds Γ in Γ_∞. Applying α levelwise induces an injective endomorphism σ of Γ_∞. It is surjective, hence an automorphism, because an element at level k equals σ of the same element at level $k+1$. Put $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$, with stable letter t implementing σ; then $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$.

Now write $X = G/\Gamma$ for the coset space of the level-zero copy, and let $\mathrm{Cl}(X)$ be the Clifford group on X: it is generated by a central involution ζ and involutions c_x, $x \in X$, subject to $[c_x, c_y] = \zeta$ for $x \neq y$, and every permutation of X acts on it by permuting the c_x and fixing ζ (Construction 8.1). Let G act in this way and map c to c_Γ. Then

$$\hat{c} \longmapsto c_{t\Gamma}, \qquad a\hat{c}a^{-1} \longmapsto c_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \longmapsto (c_{t\Gamma} c_{at\Gamma})^2 = \zeta \neq 1.$$

The sign ζ is central, so the map respects every defining relation. It follows both that $w \neq 1$ and that the canonical map of Γ is injective.

Thus $w = u^2$ is a nontrivial central involution, and by Theorem 4.2 every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps w to the identity. Since $w \neq 1$, every such homomorphism has nontrivial kernel. $\square$

Finite presentability of Γ is used only to make $E(\Gamma, \alpha, a)$ finitely presented; for the rest, Γ may be any property-(T) group with a proper injective endomorphism.

6. A finitely presented non-MF group

Specialize Theorem 5.1 to

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I). \tag{5}$$

The affine group Γ is finitely presented and has property (T) [BHV, Example 1.7.4(i)], and Section 7 identifies it with the presentation used below. The endomorphism α is injective because doubling is injective on $\mathbb{Z}^3$, and it is not surjective because $e_1 \notin 2\mathbb{Z}^3$. The same observation gives $a \notin \alpha(\Gamma)$. Hence Theorem 5.1 applies. Relative to Γ, the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \;\middle|\; \begin{array}{c} t\gamma t^{-1} = \alpha(\gamma), \quad [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, \qquad [w, g] = 1 \end{array} \right\rangle,$$

where S_Γ is a finite generating set of Γ, g runs over the generators, and $w = [\,tct^{-1},\, a\,(tct^{-1})\,a^{-1}]$. Substituting a standard presentation of the affine group in the generators v_1, v_2, v_3 for the translations and x, y, z for the linear part yields the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$\mathcal{R} = \left\langle x, y, z \;\middle|\; \begin{array}{c} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle, \tag{6}$$

and the *base*

(7)

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \;\middle|\; \begin{array}{c} \text{the eight relations in (6)}, \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3,\ xv_2x^{-1} = v_1,\ xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1,\ yv_2y^{-1} = v_2^{-1}v_3,\ yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1},\ zv_2z^{-1} = v_1v_3^{-1},\ zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write $\iota \colon \mathcal{B} \to E$ for the homomorphism induced by the first six generators of the presentation below.

Definition 6.1. Use the six base generators and the relations displayed in (7). Define

(8)
$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \;\middle|\; \begin{array}{c} \text{(B)} \quad \text{all relations in (7)}, \\ \text{(T)} \quad tv_it^{-1} = v_i^2 \ (1 \le i \le 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C)} \quad c^2 = 1, \\ [c, g] = 1 \ \ (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W)} \quad [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$

where w abbreviates the word $\left[tct^{-1},\, v_1(tct^{-1})v_1^{-1}\right]$. This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Centrality of w together with $c^2 = 1$ implies $w^2 = 1$ by (4). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \qquad [c, \iota(\mathcal{B})] = 1.$$

Family (B) presents the Kazhdan base, family (T) imposes $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$, family (C) makes c an involution centralizing the base, and family (W) makes w central; these are the hypotheses of Theorem 4.2. Families (T) and (C) imply that $\hat{c} = tct^{-1}$ commutes with $t\iota(\mathcal{B})t^{-1}$. The element $\iota(v_1)$ lies outside that conjugate subgroup, and the commutator $[\,\hat{c}, \iota(v_1)]$ is nontrivial (Proposition 8.3 and Lemma 6.2); Figure 1 depicts this configuration.

A Tietze reduction removes two generators and nine relators (Section 7).

Lemma 6.2 (the distinguished word is a square)**.** *Let H_1 be a group and $x, y \in H_1$ with $x^2 = 1$. Then*

$$[\,x,\ y\,x\,y^{-1}] \;=\; [\,x, y\,]^2.$$

Proof. Put $z = yxy^{-1}$, so $z^2 = yx^2y^{-1} = 1$; thus $z^{-1} = z$ and $x^{-1} = x$. Then $[x, y] = xyx^{-1}y^{-1} = x\,(yxy^{-1}) = xz$, and

$$[x, y]^2 = (xz)^2 = xz\,xz = x\,z\,x^{-1}z^{-1} = [x, z]. \qquad \square$$

Proof of Theorem A. The group E is the specialization of Theorem 5.1 to the data (5), and $w \neq 1$ by Proposition 8.3. We check the hypotheses of the central involution obstruction on the presentation (8) itself, so that the argument reads off the displayed relations.

Let (d_n) be given, write $\mathcal{Q} = \mathcal{Q}((d_n))$, and let $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Apply the central involution obstruction (Theorem 4.2) with $\Gamma = \mathcal{B}$, the canonical homomorphism ι, the elements $t, c \in E$, and $a = v_1$. Proposition 7.1 gives property (T) of $\mathcal{B}$, while families (T) and (C)

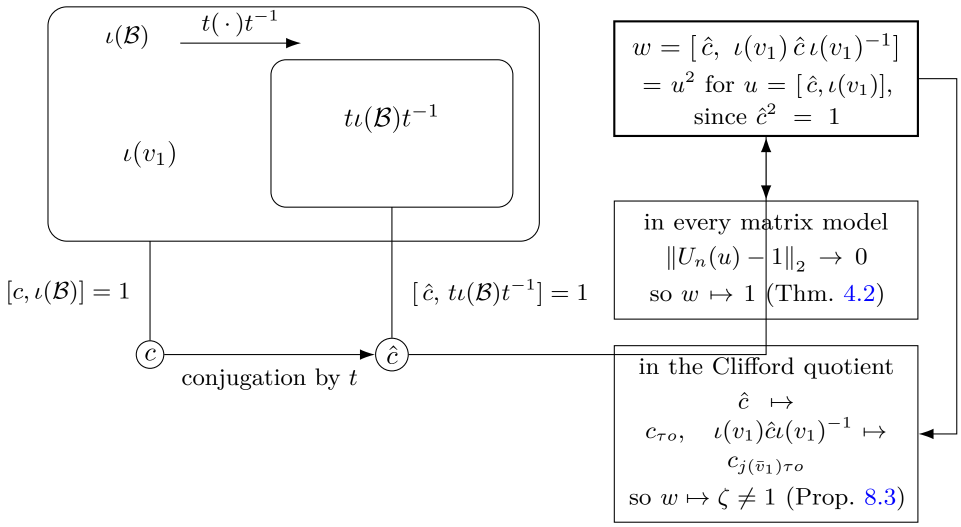

Figure 1: The distinguished word has two values. The relations of (8) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, hence $[\,\hat{c},\ t\iota(\mathcal{B})t^{-1}] = 1$ for $\hat{c} = tct^{-1}$. Every homomorphism to $\mathcal{U}(\mathcal{Q})$ sends w to the identity, while the Clifford quotient of Section 8 sends it to the nontrivial sign ζ; the two together are Theorem A.

give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, respectively. In E we have $\hat{c}^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 6.2 with $a_1 = \iota(v_1)$ gives

$$w = [\,\hat{c},\ \iota(v_1)\,\hat{c}\,\iota(v_1)^{-1}] = [\,\hat{c}, \iota(v_1)\,]^2 = u^2, \qquad u = [\,\hat{c}, \iota(v_1)\,].$$

The defining relations make w central, and $c^2 = 1$ then implies $w^2 = 1$ by (4). Hence $\Theta(w) = 1$.

Every homomorphism $E \to \mathcal{U}(\mathcal{Q})$ maps the nontrivial element w to 1, so none is injective and E is not MF. The canonical maps $E \to \mathcal{U}(C^*_{\mathrm{r}}(E))$ and $E \to \mathcal{U}(C^*_{\max}(E))$ are injective, the first because $g \mapsto \lambda_g$ is and the second because the left regular representation factors through $C^*_{\max}(E)$. If $e\colon A \to \mathcal{Q}$ is an injective, possibly nonunital, $*$-homomorphism from a unital algebra, then $u \mapsto e(u) + 1 - e(1)$ is an injective homomorphism $\mathcal{U}(A) \to \mathcal{U}(\mathcal{Q})$. Hence an MF embedding of either $C^*_{\mathrm{r}}(E)$ or $C^*_{\max}(E)$ would give an injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$, a contradiction. □

Corollary 6.3 (uniform finite operator-norm obstruction)**.** *There exist $\delta > 0$ and a finite set $F_0 \subset E$ such that, for every $d \geq 1$, every map $\phi\colon E \to \mathcal{U}(d)$ satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \qquad (g, h \in F_0)$$

satisfies $\|\phi(w) - 1\| < 1$.

Proof. Fix an exhaustion $F_1 \subseteq F_2 \subseteq \cdots$ of E by finite sets. If no such pair (δ, F_0) exists, then for each n there is a unitary-valued ϕ_n with multiplicative defect at most $1/n$ on F_n and $\|\phi_n(w) - 1\| \geq 1$. The sequence (ϕ_n) is an operator-norm asymptotic representation of E, so by Lemma 2.2 its class is a homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$, and $\|\Theta(w) - 1\| \geq 1$ gives $\Theta(w) \neq 1$, contradicting Theorem A. □

Since E is finitely presented, the forty-one relators of (8) serve as the test set, and the hypothesis becomes a condition on an eight-tuple of unitaries.

Corollary 6.4 (a uniform obstruction on the presentation)**.** *There is a $\delta > 0$ with the following property. For every $d \geq 1$ and every eight-tuple $V_1, \dots, V_8 \in \mathcal{U}(d)$, if*

$$\|r(V_1, \dots, V_8) - 1\| \leq \delta \qquad \textit{for each of the forty-one relators } r \textit{ of } (8),$$

then $\|w(V_1, \dots, V_8) - 1\| < 1$.

Proof. If no such δ exists, then for every n there are a dimension d_n and a tuple $V^{(n)}$ satisfying every relator to within $1/n$ and with $\left\|w(V^{(n)}) - 1\right\| \geq 1$. Each tuple defines a homomorphism $\varphi_n\colon F_8 \to \mathcal{U}(d_n)$, since F_8 is free, and by Lemma 2.2 the classes define a homomorphism $\Phi\colon F_8 \to \mathcal{U}(\mathcal{Q}((d_n)))$. Each of the forty-one relators lies in $\ker \Phi$, its images converging to 1 in operator norm, so Φ factors as $\Theta \circ \pi$ through the quotient map $\pi\colon F_8 \to E$. Theorem A gives $\Theta(w) = 1$, that is $\left\|w(V^{(n)}) - 1\right\| \to 0$, contradicting the lower bound 1. □

The compactness argument gives no bound for δ or $|F_0|$; an effective modulus would need an effective Kazhdan constant for the base together with effective versions of the compression estimates.

7. The base and a smaller presentation

The base $\mathcal{B}$ of (7) is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, for which property (T) is classical [BHV, Example 1.7.4(i)].

A six-generator presentation. The group E is generated by the six elements v_1, x, y, z, t, c, and has a Tietze-equivalent presentation on them with thirty-two relators. In (8), eliminate $v_3 = xv_1x^{-1}$ and $v_2 = x^2v_1x^{-2}$ by Tietze substitutions and delete the two defining relations; the remaining x-action relation follows from $x^3 = 1$, and the stable-letter, c-commutation, and centrality relations for v_2, v_3 follow from those for v_1 together with $[t, x] = 1$ and $x^3 = 1$. Deleting these nine leaves thirty-two relators on v_1, x, y, z, t, c.

After these Tietze transformations, the remaining relations still imply $[\,\hat{c},\, t\iota(\mathcal{B})t^{-1}] = 1$ for $\hat{c} = tct^{-1}$, and the distinguished word $[\,\hat{c},\ \iota(v_1)\,\hat{c}\,\iota(v_1)^{-1}]$, computed in the same group, is unchanged.

7.1. **Property (T) of the base.**

Proposition 7.1. *The twenty-relator group $\mathcal{B}$ has property* (T)*, in both the real-orthogonal and complex-unitary formulations.*

Proof. By Remark 7.2 the presentation (7) defines $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) [BHV, Example 1.7.4(i)]. For an orthogonal representation ρ on a real Hilbert space $H_{\mathbb{R}}$, the complexification $H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ carries the unitary representation $\rho \otimes 1$, whose invariant vectors and displacement norms are those of ρ in each of the two real coordinates; a Kazhdan pair for the unitary formulation is therefore one for the orthogonal formulation. □

Remark 7.2 (the classical identification)*.* The eight relations (6) present $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2]; the corresponding matrices are also recorded in [CLV]. Hence $\mathcal{B} \cong \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, and the surjection $\mathcal{B} \to \bar{\Gamma}$ of Lemma 8.2 onto the matrix realization is an isomorphism.

8. Nontriviality of w

Construction 8.1 (Clifford groups)**.** For a set X, use the real Clifford-algebra convention generated by $(c_x)_{x\in X}$ with $c_x^2 = 1$ and $c_x c_y = -c_y c_x$ for $x \neq y$, and let $\mathrm{Cl}(X)$ be the subgroup of units generated by -1 and the c_x. Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \qquad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Thus $\mathrm{Cl}(X)$ is a central extension of the elementary abelian 2-group on X, with central kernel generated by ζ. For distinct $x, y \in X$, the commutator $[c_x, c_y]$ equals ζ. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ.

Lemma 8.2 (linear model)**.** *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M,u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$x \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), \qquad y \mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right),$$

$$z \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right).$$

Let $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2,2,2,1)$. The six matrices satisfy all twenty relators of $\mathcal{B}$, and

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z. Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar{\alpha})$*.*

Proof. Each of the twenty relators reduces to an equality between explicit 4×4 rational matrices. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar\Gamma \leq \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar v_1 D$ has entry $1/2$ in position $(1,4)$, so it is not in $\bar\Gamma$. This proves the last assertion. $\square$

Form the direct limit $T = \varinjlim(\bar\Gamma,\bar\alpha)$ and let τ denote its shift automorphism. Set $\Sigma = T \rtimes \mathbb{Z}$. This is the group G used in Theorem 5.1 for the data (5). Let $j\colon \bar\Gamma \hookrightarrow \Sigma$ be the level-zero map. Put $X = \Sigma/j(\bar\Gamma)$, with base point $o = j(\bar\Gamma)$. Since $D^{-1}\bar v_1 D \notin \bar\Gamma$ (Lemma 8.2), we have $\tau o \neq j(\bar v_1)\tau o$, and the Clifford generators at these two points anticommute.

Proposition 8.3. *$w \neq 1$ in E.*

Proof. Form $W := \mathrm{Cl}(X) \rtimes \Sigma$ as in Construction 8.1. Define a map on the generators of (8) by

$$g \mapsto j(\bar g) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}),$$
$$t \mapsto \tau, \qquad c \mapsto c_o.$$

The exact checks in Lemma 8.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o, so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \qquad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar v_1)\tau o},$$

and the two points are distinct by Lemma 8.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar v_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W, the remaining relations $[w,g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \to W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E. $\square$

9. Consequences of the example

Throughout, $\hat c = tct^{-1}$ and $u = [\hat c, \iota(v_1)]$, so that $w = u^2$. For a countable group G, write $\mathrm{Res}_{\mathrm{MF}}(G)$ for its *MF residual*: the intersection of the kernels of all its homomorphisms to $\mathcal{U}(\mathcal{Q})$ (Definition S6.1). Proposition S6.3 shows that G is MF exactly when this subgroup is trivial. By Theorem A, $1 \neq w \in \mathrm{Res}_{\mathrm{MF}}(E)$.

The quotient by the central involution.

Theorem B (the quotient of E by $\langle w\rangle$)**.** *The quotient $E/\langle w\rangle$ is not MF. The image of $[\iota(v_1), \hat c] = u^{-1}$ is nontrivial in $E/\langle w\rangle$ and lies in $\mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle)$; consequently $\{1, w\}$ is a proper subgroup of $\mathrm{Res}_{\mathrm{MF}}(E)$.*

By Theorem S7.3, if an involution commutes with the conjugate copy of a Kazhdan subgroup and its conjugates under the original subgroup commute pairwise, then every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps all of those conjugates to the same element; Section S7 proves this and applies it here. In E the pairwise commutators of those conjugates of $\hat{c}$ lie in $\langle w \rangle$, so in $E/\langle w \rangle$ the element $\hat{c}$ satisfies these hypotheses, and every homomorphism $E/\langle w \rangle \to \mathcal{U}(\mathcal{Q})$ maps u to 1. Thus $E/\langle w \rangle$ is a second finitely presented non-MF group.

A stably finite non-MF group algebra. Every MF algebra is stably finite [BK]; the converse fails. The canonical trace on $C^*_{\mathrm{r}}(E)$ is faithful, so $C^*_{\mathrm{r}}(E)$ is stably finite (Lemma S8.1). Abstract examples follow from the failure of the Connes embedding problem [FGH]; the reduced algebra of the finitely presented group E is an explicit one (Theorem C). For $C^*_{\mathrm{r}}(\mathrm{SL}_3(\mathbb{Z}))$ and $C^*_{\mathrm{r}}(\mathrm{SL}_4(\mathbb{Z}))$ the MF property is undecided; only the stronger PMF property is known to fail, for $\mathrm{SL}_4(\mathbb{Z})$ [MdlS].

Theorem C (the reduced group algebra of E)**.** *The algebra $C^*_{\mathrm{r}}(E)$ is unital and separable, has a faithful tracial state, and is therefore stably finite, but it is not an MF algebra.*

Proof. $C^*_{\mathrm{r}}(E)$ is unital and separable because E is countable, and by Lemma S8.1 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A. □

The block normal form of Section S8 shows in addition that E, and hence $C^*_{\mathrm{r}}(E)$, is exact. The canonical map $\mathcal{B} \to E$ is injective (Section 8), so E contains an infinite Kazhdan group and is nonamenable. By Lance's theorem, nonamenability of a discrete group is equivalent to nonnuclearity of its reduced group C^*-algebra, so $C^*_{\mathrm{r}}(E)$ is not nuclear [Lan]. In the direction used here, nuclearity is read as the completely positive approximation property: a discrete group whose reduced algebra has that property carries an invariant mean, and E does not. So $C^*_{\mathrm{r}}(E)$ says nothing about the nuclear form of the Blackadar–Kirchberg problem, whether every stably finite separable nuclear C^*-algebra is quasidiagonal.

The maximal algebra behaves differently: $C^*_{\max}(E)$ is not stably finite and has no faithful tracial state. This holds for every group containing a property-(T) subgroup properly conjugated into itself, and comes from a proper isometry built directly from the Kazhdan projection (Proposition S11.3).

A sofic example.

Theorem D (a finitely presented sofic non-MF group)**.** *The group E of Definition 6.1 is finitely presented, sofic, hyperlinear, and non-MF. It is not LEF and therefore is not residually finite.*

Let E_0 be the kernel of the exponent sum in t. For every finite subset of E_0, the T-coordinates lie in a single level of the ascending chain and the lamp coordinates involve finitely many conjugates of c; the subgroup they generate embeds in a residually finite group. Hence E_0 is LEF (Section S8.3). Since E is a split extension of E_0 by $\mathbb{Z}$, it is sofic by the Elek–Szabó amenable-extension theorem [ES06]. This extension does not preserve LEF: LEF groups are MF [CDE], so E_0 is LEF whereas E is not.

The MF property is closed under neither extensions nor quotients: a Clifford quotient of E is a non-MF extension of an MF group by an MF group, and the MF group F_8 surjects onto E (Section S8, Corollary S6.8).

A hyperlinear non-MF trace. We use Shulman's terminology [ShT]. Hyperlinear and MF traces are defined by the same pointwise bounded, trace-correct matrix models, with the algebraic defects measured in normalized Hilbert–Schmidt norm and operator norm, respectively. The approximating maps themselves need not be linear, positive, unital, or completely positive. Every MF trace is hyperlinear; Shulman asks whether the converse holds.

Lemma 9.1. *Let G be a countable group and let τ_G be the canonical trace on $C^*_{\max}(G)$, determined by $\tau_G(u_g) = 1$ for $g = 1$ and $\tau_G(u_g) = 0$ otherwise. If τ_G is an MF trace, then G is MF.*

Proof. Let $\varphi_n \colon C^*_{\max}(G) \to M_{k_n}(\mathbb{C})$ witness that τ_G is MF and put $\mathcal{Q} = \prod_n M_{k_n}(\mathbb{C}) / \bigoplus_n M_{k_n}(\mathbb{C})$, the quotient of the bounded product by the operator-norm c_0 ideal. Since the models are pointwise bounded, $(\varphi_n(a))_n$ lies in the bounded product for every a, while the operator-norm defect conditions imply that $\Phi(a) = [(\varphi_n(a))_n]$ is a $*$-homomorphism $\Phi \colon C^*_{\max}(G) \to \mathcal{Q}$. Set $p = \Phi(1)$, a projection with $\Phi(C^*_{\max}(G)) \subseteq p\mathcal{Q}p$. Then $\Theta(g) = \Phi(u_g) + (1-p)$ defines a homomorphism $\Theta \colon G \to \mathcal{U}(\mathcal{Q})$. We claim that Θ is injective. If $g \neq 1$ and $\Theta(g) = 1$, then $\Phi(u_g) = \Phi(1)$, so $\|\varphi_n(u_g) - \varphi_n(1)\| \to 0$; the normalized trace is contractive for the operator norm, whereas trace-correctness gives $\mathrm{tr}_{k_n}(\varphi_n(u_g)) \to 0$ and $\mathrm{tr}_{k_n}(\varphi_n(1)) \to 1$, a contradiction. Thus G embeds in the unitary group of a norm matrix corona and is MF (Proposition 2.3). $\square$

Schafhauser [Sc, Definition 1.1 and Proposition 2.2] uses a stronger notion of MF group and relates it to the canonical trace on $C^*_{\mathrm{r}}(G)$; Lemma 9.1 concerns instead the Carrión–Dadarlat–Eckhardt notion used here and the canonical trace on $C^*_{\max}(G)$.

Theorem E (a hyperlinear non-MF trace)**.** *The canonical trace τ_E on $C^*_{\max}(E)$ is hyperlinear but not MF. Consequently, there is a separable unital C^*-algebra carrying a hyperlinear non-MF trace. In particular, this gives a negative answer to Shulman's question whether every hyperlinear trace is MF* [ShT].

Proof. By Theorem D, E is sofic. Let $\sigma_n \colon E \to \mathrm{Sym}(d_n)$ be a sofic approximation and $U_n(g)$ the associated permutation unitaries. For a free ultrafilter ω, let

$$\oplus_{2,\omega} M_{d_n}(\mathbb{C}) = \{(x_n) : \lim_{\omega} \|x_n\|_{2,\mathrm{tr}_{d_n}} = 0\}$$

inside the bounded product, and write $\mathrm{tr}_\omega([(x_n)]) = \lim_\omega \mathrm{tr}_{d_n}(x_n)$ for the induced trace on the tracial ultraproduct. The classes $[(U_n(g))_n]$ define a homomorphism

$$E \longrightarrow \mathcal{U}\left(\prod_n M_{d_n}(\mathbb{C}) / \oplus_{2,\omega} M_{d_n}(\mathbb{C})\right).$$

Moreover, asymptotic freeness of the sofic approximation gives

$$\mathrm{tr}_\omega([(U_n(g))_n]) = \delta_{g,1},$$

because the normalized trace of a permutation matrix is the proportion of its fixed points. By the universal property of $C^*_{\max}(E)$ this extends to a $*$-homomorphism π with $\tau_E = \mathrm{tr}_\omega \circ \pi$, so τ_E is hyperlinear by Shulman's tracial-ultraproduct characterization [ShT].

If τ_E were an MF trace, Lemma 9.1 would imply that E is MF, contradicting Theorem A. Separability of $C^*_{\max}(E)$ follows from countability of E. $\square$

Supplementary material

Sections S1–S3 contain the structural and analytic tools quoted by the main text: the MF equivalences, weighted asymptotic-commutant invariance, and the general obstruction criteria. Sections S4–S8 study the group E itself and its MF residual, and the remaining sections collect further consequences, the limits of the method, and open questions.

S1. The MF equivalences

An *approximate unitary representation* for a countable group G, a finite set $F \subseteq G$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$ is a map $V : G \to \mathcal{U}(r)$, for some positive dimension r, such that

$$\|V(g)V(h) - V(gh)\| \le \varepsilon \quad (g, h \in F),$$
$$\|V(g) - V(h)\| \ge \delta \quad (g \ne h \in F).$$

The local formulation in Proposition 2.3 asks for such a model with $\delta = 1$ for every finite F and every $\varepsilon > 0$.

Proof of Proposition 2.3. Lemma 2.2 identifies $\prod_n \mathcal{U}(d_n)/\mathcal{N}$ with $\mathcal{U}(\mathcal{Q})$, which is the first equivalence once the dimension convention is normalized.

Dimension normalization. Given positive dimensions d_n, replace the nth sequence of unitary lifts by

$$\widetilde{V}_n(g) = I_{\mathbb{C}^{d_0}} \oplus \cdots \oplus I_{\mathbb{C}^{d_{n-1}}} \oplus V_n(g)$$

on $\mathbb{C}^{d_0} \oplus \cdots \oplus \mathbb{C}^{d_n}$, of strictly increasing dimension $D_n = d_0 + \cdots + d_n$. Block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements, so for all g, h

$$\left\|\widetilde{V}_n(g)\widetilde{V}_n(h) - \widetilde{V}_n(gh)\right\| = \|V_n(g)V_n(h) - V_n(gh)\|,$$
$$\left\|\widetilde{V}_n(g) - \widetilde{V}_n(h)\right\| = \|V_n(g) - V_n(h)\|.$$

The defect and separation data are unchanged. Hence the padded lifts define a homomorphism to $\mathcal{U}(\mathcal{Q}((D_n)))$, and it separates exactly the same group elements as the original homomorphism. The converse is immediate.

Suppose first that $\rho : G \to \mathcal{U}(\mathcal{Q}((d_n)))$ is injective, and choose unitary lifts $u_n(g)$ of $\rho(g)$. Fix a finite set $F \subseteq G$ and $\varepsilon > 0$.

Tensor-power amplification. If $u, v \in \mathcal{U}(d)$ satisfy $\|u - v\| \ge \delta$ and $8 < N\delta^2$, then $\|u^{\otimes p} - v^{\otimes p}\| \ge 1$ for some $p \le N$. Indeed, put $z = u^*v$, so that $\|u - v\| = \|1 - z\|$ and z has a spectral value $e^{2\pi i\alpha}$ with $|1 - e^{2\pi i\alpha}| = 2\sin(\pi\alpha') \ge \delta$, where $\alpha' = \min(\{\alpha\}, 1 - \{\alpha\})$. Since $(u^{\otimes p})^* v^{\otimes p} = z^{\otimes p}$ has $e^{2\pi i p\alpha}$ in its spectrum,

$$\left\|u^{\otimes p} - v^{\otimes p}\right\| \;\ge\; \left|1 - e^{2\pi i p\alpha}\right| \;=\; 2\sin\bigl(\pi \min(\{p\alpha\}, 1 - \{p\alpha\})\bigr).$$

This is at least 1 whenever $\{p\alpha\} \in [1/6, 5/6]$. The bound $2\sin(\pi\alpha') \ge \delta$ gives $\alpha' \ge \delta/(2\pi)$; consecutive multiples of α move by α' modulo 1, so some $p \le \lceil 1/(6\alpha')\rceil \le \pi/(3\delta) + 1 < N$ lands in that interval.

Building the local model. For each ordered pair $g \neq h$ in F, injectivity says that $(u_n(g)^{-1}u_n(h))_n$ does not converge to the identity in operator norm. Hence there are $\delta_{g,h} > 0$ and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer $N_{g,h}$ with $8 < N_{g,h}\delta_{g,h}^2$. Because ρ is a homomorphism into the quotient, all multiplicative defects for pairs in F are arbitrarily small at every sufficiently large coordinate. For each $g \neq h$, choose a coordinate with the displayed separation and with every required defect at most $\varepsilon/N_{g,h}$. Tensor-power amplification gives $p \leq N_{g,h}$ for which the pth tensor power separates the designated pair by at least 1; the tensor-product telescoping identity bounds each new defect by $p\varepsilon/N_{g,h} \leq \varepsilon$.

Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in F is separated by at least 1 and every required product has defect at most ε. This gives the normalized approximate representation. If $|F| \leq 1$, there are no pairs to separate, and the constant map to $1 \in \mathcal{U}(1)$ serves.

Conversely, suppose approximate unitary representations with separation constant 1 exist for all finite sets and tolerances. Choose models along an exhaustion with errors tending to zero. For each fixed pair g, h, the multiplicative defect tends to 0, and for each fixed pair $g \neq h$, the separation is eventually at least 1. Hence the classes define an injective homomorphism into $\prod_n \mathcal{U}(r_n)/\mathcal{N}$, with no condition on the dimension sequence. Composing it with the isomorphism $\kappa_{(r_n)}$ of Lemma 2.2 gives an injective homomorphism $G \to \mathcal{U}(\mathcal{Q}((r_n)))$, which is the group-MF definition.

$\square$

S2. Weighted asymptotic-commutant invariance

Theorem 3.4 holds with an arbitrary weight ν_n in place of the dimension.

Theorem S2.1 (one-sided conjugation at an arbitrary weight)**.** *Let Γ, H, ι, and s be as in Theorem 3.4, let $U_n \colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation, let (ν_n) be nonnegative weights, and let $x_n \in M_{d_n}(\mathbb{C})$ be matrices for which there is C with $\operatorname{Tr}(x_n^* x_n) \le C\,\nu_n$ for all n and, for every $\gamma \in \Gamma$ and every $\varepsilon > 0$, eventually*

$$\operatorname{Tr}\left|x_n - U_n(\iota(\gamma))\,x_n\,U_n(\iota(\gamma))^*\right|^2 \le \varepsilon\,\nu_n.$$

Then $U_n(s)\,x_n U_n(s)^*$ and $U_n(s)^*\,x_n U_n(s)$ satisfy both conditions as well.

Proof. Equip the nth coordinate space with the inner product

$$\langle x, y\rangle_n = \begin{cases} \operatorname{Tr}(y^*x)/\nu_n, & \nu_n > 0, \\ \operatorname{Tr}(y^*x), & \nu_n = 0. \end{cases}$$

If $\nu_n = 0$, the bound $\operatorname{Tr}(x_n^* x_n) \le C\nu_n$ forces $x_n = 0$, so the harmless choice of the ordinary Hilbert–Schmidt inner product on that coordinate does not affect the distinguished vector. Conjugation by a unitary is unitary for every one of these inner products. The bound makes (x_n) a bounded ultraproduct vector, and the adjoint actions remain operator-norm almost multiplicative because the scalar normalization of the Hilbert norm does not enter the operator norm. Thus the ultraproduct proof of Theorem 3.4 applies with these weighted coordinate Hilbert spaces. Finiteness of the norm ultraproduct again turns $P \le VPV^*$ into $P = VPV^*$, which gives both conjugation directions. □

Taking $\nu_n = k_n$, the rank of the projection lift in the proof of Theorem S7.1, gives the normalization used there.

S3. Obstructions from a one-sided conjugation datum

Theorem 3.4 puts the normal closure N_{conj} of the commutators $[tct^{-1}, \iota(\lambda)]$ inside the 2-norm kernel of every operator-norm asymptotic representation of H. Each result below combines that fact with Lemma 4.1 for a different choice of corner, on which the subgroup at issue stays a definite 2-norm distance from the identity, and so lands in the kernel of every homomorphism $H \to \mathcal{U}(\mathcal{Q})$. For a finite normal subgroup of N_{conj} the corner is the complement of a projection obtained from its averaging operators (Theorem S3.3); the central involution of Section 4 is the case $F = \{1, w\}$. For a normal property-(T) subgroup — possibly infinite, noncentral and torsion-free — the corner is a spectral projection of a Kazhdan average (Theorems S3.6 and S3.7). Corollary S3.8 replaces N_{conj} by the subgroup $\mathfrak{D}(H, L)$ of (9), which depends on L alone and not on a choice of t and c.

Definition S3.1 (Kazhdan conjugation datum)**.** Let H be a countable group. A *Kazhdan conjugation datum* in H is a tuple (Λ, ι, t, c), where Λ is a countable group with property (T), $\iota\colon \Lambda \to H$ is a group homomorphism, possibly with nontrivial kernel, and $t, c \in H$, subject to

(M1) $t\,\iota(\lambda)\,t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;

(M2) $c\,\iota(\lambda) = \iota(\lambda)\,c$ for every $\lambda \in \Lambda$.

Write $\hat{c} := tct^{-1}$. The datum determines the normal closure

$$N_{\mathrm{conj}} \;=\; \big\langle\!\big\langle\, [\,\hat{c}, \iota(\lambda)\,] \;:\; \lambda \in \Lambda \,\big\rangle\!\big\rangle_H \;\trianglelefteq\; H.$$

For a finitely generated Λ, conditions (M1)–(M2) are imposed by finitely many relations once generating sets are fixed. Conditions (M1)–(M2) imply that c commutes with $\iota(\Lambda)$ and that $\hat{c}$ commutes with the subgroup $t\iota(\Lambda)t^{-1}$.

Definition S3.2 (universal 2-norm triviality)**.** Let $U = (U_n)$ be an operator-norm asymptotic representation of a countable group H and let ω be an ultrafilter refining the cofinite filter. Write

$$K_2^\omega(U) \;=\; \{g \in H :\; \lim_\omega \|U_n(g) - 1\|_2 = 0\},$$

the kernel of the homomorphism induced by U into the tracial ultraproduct of the matrix blocks along ω, and hence a normal subgroup of H. An element g is *universally* 2*-norm trivial* if $g \in K_2^\omega(U)$ for every such U and every such ω. If $\|U_n(g) - 1\|_2 \to 0$ along the full sequence, then $g \in K_2^\omega(U)$ for every such ω.

Theorem S3.3 (finite-normal obstruction criterion)**.** *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c), and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\mathrm{conj}}$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Proof. *Universal* 2*-norm triviality of N_{conj}.* Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H (for instance, coordinate unitary lifts of a homomorphism to $\mathcal{U}(\mathcal{Q})$), write $U_{g,n} = U_n(g)$, and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \longrightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. Since c centralizes $\iota(\Lambda)$, the sequence $(U_{c,n})$ belongs to the asymptotic commutant of $\iota(\Lambda)$ in normalized Hilbert–Schmidt norm. Theorem 3.4, applied to $x_n = U_{c,n}$, shows that $(U_{\hat{c},n})$ for $\hat{c} = tct^{-1}$ lies in the same asymptotic commutant. Therefore $[\hat{c}, \iota(\lambda)] \in K_2$ for every $\lambda \in \Lambda$; since K_2 is normal, it contains the normal closure of these commutators: $N_{\mathrm{conj}} \subseteq K_2$, and so $F \subseteq K_2$. Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of N_{conj} is universally 2-norm trivial (Definition S3.2).

The coordinate averaging corner. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F; choose coordinate unitary lifts $(V_{g,n})$ of Θ and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and F is finite, (p_n) is asymptotically self-adjoint and idempotent. Since F is normal, (p_n) asymptotically commutes with every lift of $\Theta(H)$. Continuous functional calculus applied to the Hermitian parts yields genuine projections at operator-norm distance $o(1)$ from p_n, and these projections are asymptotically central as well. Let q_n be their complements. Nontriviality of Θ on F forces $q_n \neq 0$ along infinitely many coordinates — otherwise the averages converge to 1 and $\Theta|_F$ is trivial. Retain exactly those coordinates and relabel them by $\mathbb{N}$. Lemma 4.1 gives an operator-norm asymptotic representation $(W_{g,n})$ of H on the corners $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$, $r_n = \operatorname{rank} q_n > 0$. Since q_n is the complement of a projection within $o(1)$ of p_n, the finite average vanishes on the corner:

$$(*) \qquad \Big\| \sum_{f \in F} W_{f,n} \Big\| \longrightarrow 0 \quad \text{in operator norm.}$$

The contradiction. Since $F \subseteq N_{\text{conj}}$ and $(W_{g,n})$ is itself an operator-norm asymptotic representation of H, every $f \in F$ satisfies $\|W_{f,n} - 1\|_2 \to 0$. Because F is finite, these convergences are uniform over $f \in F$, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F| 1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts $(*)$, because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ is trivial on F. $\square$

The same conclusion holds with N_{conj} replaced by any normal subgroup all of whose elements lie in $K_2^\omega(U)$ for every operator-norm asymptotic representation U of H and every ω; Theorem S3.5 is that statement for $\mathfrak{D}(H, L)$.

The central involution obstruction (Theorem 4.2) is the case $F = \{1, w\}$, where $p_n = \frac{1}{2}(1 + V_{w,n})$; in the corona corner the image of w is -1 exactly, while the finite-stage compressions satisfy $\|W_{w,n} + 1\| \to 0$.

S3.1. **An intrinsic form.** For a subgroup $L \leq H$, let

$$P_L = \{s \in H : sLs^{-1} \subseteq L\},$$

the *compression semigroup* of L in the terminology of Kun–Thom [KT26], and put $G^+(L) = \langle P_L \rangle$. Then set

$$(9) \qquad \mathfrak{D}(H, L) = \Big\langle\!\Big\langle [gzg^{-1}, \ell] : g \in G^+(L),\ z \in C_H(L),\ \ell \in L \Big\rangle\!\Big\rangle_H,$$

which depends only on the pair (H, L), not on a choice of datum.

Corollary S3.4 (transport by the compression semigroup)**.** *Let (r_n) and $V_{g,n} \in \mathcal{U}(r_n)$ be an operator-norm asymptotic representation of H, let Λ have property* (T)*, let $\iota \colon \Lambda \to H$ be a homomorphism, and put $L = \iota(\Lambda)$. Suppose* *that $(x_n \in M_{r_n}(\mathbb{C}))$ is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of L:*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \qquad (\lambda \in \Lambda).$$

Then for every $g \in G^+(L)$, conjugation and inverse conjugation both preserve that asymptotic commutant. Explicitly, both

$$V_{g,n} x_n V_{g,n}^* \quad \textit{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every $\lambda \in \Lambda$.

Proof. For a one-sided conjugator s, Theorem 3.4 gives the conclusion for $V_{s,n} x_n V_{s,n}^*$. The equality of the fixed-space projection with its conjugate established in that proof also gives invariance under the inverse conjugation, hence the conclusion for $V_{s,n}^* x_n V_{s,n}$. The set of $g \in H$ for which conjugation by $V_{g,n}$ and by $V_{g,n}^*$ both preserve the asymptotic commutant is a subgroup, and it contains every one-sided conjugator; hence it contains $G^+(L)$. □

Theorem S3.5 (the subgroup $\mathfrak{D}$ and the MF residual)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \le \mathfrak{D}(H, L)$, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps every element of F to the identity.*

Proof. Let $(U_{h,n})$ be coordinate lifts of an operator-norm asymptotic representation of H and put $K_2 = \{h : \|U_{h,n} - 1\|_2 \to 0\}$, a normal subgroup of H. If $z \in C_H(L)$, then $(U_{z,n})$ lies in the asymptotic commutant of L, and by Corollary S3.4 so does $(U_{gzg^{-1},n})$ for every $g \in G^+(L)$. Hence each generator $[gzg^{-1}, \ell]$, $\ell \in L$, displayed in (9) belongs to K_2; since $K_2 \trianglelefteq H$, it contains their normal closure: $\mathfrak{D}(H, L) \le K_2$. Every element of $\mathfrak{D}(H, L)$ is therefore universally 2-norm trivial in every operator-norm asymptotic representation of H. If a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ were nontrivial on a finite normal $F \le \mathfrak{D}(H, L)$, the averaging-corner argument of Theorem S3.3 would contradict this universal 2-norm triviality. □

S3.2. **The normal-Kazhdan form of the obstruction.** A finite group has property (T), and its averaging operator is the projection onto its fixed space. The spectral projection of a Kazhdan average replaces that operator for an arbitrary normal property-(T) subgroup, which may be infinite, noncentral and torsion-free.

Theorem S3.6 (obstruction from universal 2-norm triviality)**.** *Let H be a countable group and let $D \le H$ be a subgroup every element of which is universally* 2*-norm trivial (Definition S3.2). Let $K \trianglelefteq H$ be a normal subgroup with property* (T) *such that $K \le D$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Proof. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K. Its image $\bar{H} = \Theta(H)$ is countable and, as a subgroup of $\mathcal{U}(\mathcal{Q})$, is MF by definition. Proposition 2.3, applied along an exhaustion of $\bar{H}$, gives an operator-norm asymptotic representation $(V_{g,n})$ in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models provide operator-norm separation, but at growing dimension this yields no lower bound on normalized Hilbert–Schmidt distances. Every operator-norm asymptotic representation of $\bar{H}$ composes with Θ to one of H, so every element of $\Theta(D)$ is again universally 2-norm trivial, while $\bar{K} = \Theta(K) \le \Theta(D)$ is normal, Kazhdan and nontrivial. We now work in $\bar{H}$.

The Kazhdan corner. Fix a free ultrafilter ω, let H_ω be the Hilbert-space ultraproduct of the coordinate spaces $\mathbb{C}^{d_n}$, and let $B_\omega = \prod_\omega B(\mathbb{C}^{d_n})$ act on it as in Section 3, so that the classes $\pi(g) = [V_{g,n}]_\omega$ define a unitary representation of $\bar{H}$ on H_ω. Fix a finite symmetric generating Kazhdan set $S \subseteq \bar{K}$ with constant ε_0, put $\theta = 1 - \varepsilon_0^2/(4|S|)$, and let

$$h = \frac{1}{|S|} \sum_{a \in S} \pi(a) \in B_\omega,$$

self-adjoint because S is symmetric. A unit vector ξ has $\langle h\xi, \xi\rangle = 1$ exactly when $\pi(a)\xi = \xi$ for every $a \in S$, while the Kazhdan pair bounds $\langle h\xi, \xi\rangle$ by θ on the orthogonal complement of $\operatorname{Fix} \pi(\bar{K})$. The spectrum of h therefore lies in $[-1, \theta] \cup \{1\}$, the spectral projection $P = \chi_{\{1\}}(h)$ belongs to B_ω, and its range is $\operatorname{Fix} \pi(\bar{K})$. Put $q = 1 - P$.

If $q = 0$ then $\pi(k) = 1$ for every $k \in \bar{K}$, against the operator-norm separation of a fixed nontrivial element of $\bar{K}$. For $g \in \bar{H}$ the projection $\pi(g)P\pi(g)^*$ has range $\operatorname{Fix} \pi(g\bar{K}g^{-1}) = \operatorname{Fix} \pi(\bar{K})$ by normality, so P and q commute with $\pi(\bar{H})$ exactly. Lift q to projections $q_n \in M_{d_n}(\mathbb{C})$, nonzero along ω and with $\|[q_n, V_{g,n}]_-\| \to_\omega 0$ for every $g \in \bar{H}$, and lift h to the self-adjoint coordinates

$$h_n = \frac{1}{2|S|} \sum_{a \in S} (V_{a,n} + V_{a,n}^*) \in M_{d_n}(\mathbb{C}),$$

which represent h because S is symmetric. The compression of h to the corner satisfies $qhq \le \theta q$ in B_ω, and, writing x_+ for the positive part of a self-adjoint x, the positive part is a continuous function of x and the norm ultraproduct computes it coordinatewise, so that inequality is exactly the coordinate statement

$$\left\| (q_n h_n q_n - \theta q_n)_+ \right\| \to_\omega 0. \tag{10}$$

To obtain ordinary convergence before compressing, pass to a subsequence. Enumerate $\bar{H} = \{g_1, g_2, \dots\}$. For each j the set

$$A_j = \left\{ n \ : \ q_n \neq 0, \ \ \max_{i \le j} \|[q_n, V_{g_i,n}]_-\| < 1/j, \ \ \left\| (q_n h_n q_n - \theta q_n)_+ \right\| < 1/j \right\}$$

is again a finite intersection of sets in ω — the last constraint by (10) — hence lies in ω and in particular is infinite. Choose $n_1 < n_2 < \cdots$ with $n_j \in A_j$ and relabel along $(n_j)_j$. On the relabelled sequence $q_n \neq 0$ for every n, $\|[q_n, V_{g,n}]_-\| \to 0$ in the ordinary sense for every fixed $g \in \bar{H}$, since $g = g_i$

lies in the constraint of A_j for all $j \geq i$, and the left-hand side of (10) tends to 0 in the ordinary sense as well, by the last constraint of A_j. Lemma 4.1 now gives an operator-norm asymptotic representation $(W_{g,n})$ of $\bar{H}$ on the corners $q_n M_{d_n}(\mathbb{C}) q_n$.

A 2-norm separation on the corner. Write $r_n = \operatorname{rank} q_n$ and equip the corner $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ with its renormalized trace tr_{r_n}. Applying tr_{r_n} to the ordinary form of (10) gives

$$\frac{1}{|S|} \sum_{a \in S} \operatorname{Re} \mathrm{tr}_{r_n}(q_n V_{a,n} q_n) \;=\; \mathrm{tr}_{r_n}(q_n h_n q_n) \;\leq\; \theta + o(1),$$

and polar correction moves each $q_n V_{a,n} q_n$ by $o(1)$ in operator norm, hence by $o(1)$ in tr_{r_n}, so the same bound holds with $W_{a,n}$ in its place: the average over S of the real parts of the renormalized traces of the compressed models is at most $\theta + o(1)$. For each n, at least one generator in S therefore has compressed model with real renormalized trace at most $\theta + o(1)$, and so normalized Hilbert–Schmidt distance at least $\sqrt{2(1-\theta)} - o(1)$ from the corner identity. Since S is finite, a subsequence realizes this at a single generator s_0.

The contradiction. The corner models form an operator-norm asymptotic representation of $\bar{H}$, and s_0 lies in $\bar{K} \leq \Theta(D)$. Hence s_0 is universally 2-norm trivial in the corner models: $\|W_{s_0,n} - 1\|_2 \to 0$, contradicting the positive lower bound above. $\square$

Theorem S3.7 (normal-Kazhdan obstruction)**.** *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c), and let $K \trianglelefteq H$ be a normal subgroup with property* (T) *such that $K \subseteq N_{\mathrm{conj}}$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Since finite groups have property (T), Theorem S3.3 is the finite case of Theorem S3.7.

Proof. The first step of the proof of Theorem S3.3 shows that every element of N_{conj} is universally 2-norm trivial (Definition S3.2). Theorem S3.6 with $D = N_{\mathrm{conj}}$ gives the conclusion. $\square$

Every element of $\mathfrak{D}(H, L)$ is universally 2-norm trivial, so Theorem S3.6 gives the following normal-Kazhdan analogue of Theorem S3.5:

Corollary S3.8 (normal-Kazhdan form of the intrinsic residual)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $K \trianglelefteq H$ is a normal subgroup with property* (T) *and $K \leq \mathfrak{D}(H, L)$, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps every element of K to the identity.*

Proof. The proof of Theorem S3.5 shows that every element of $\mathfrak{D}(H, L)$ is universally 2-norm trivial in every operator-norm asymptotic representation of H; apply Theorem S3.6 with $D = \mathfrak{D}(H, L)$. $\square$

The subgroup generated by *all* normal property-(T) subgroups of $\mathfrak{D}(H, L)$ therefore also lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$.

S4. The finite-dimensional obstruction

Every finite-dimensional representation of E over any field maps w to the identity (Theorem 3.3 with $H = E$). Consequently finite-dimensional linear representations do not separate the points of E, and every finite quotient maps w to the identity. Since the Clifford quotient of Section 8 sends w to a nontrivial element, that quotient is infinite. The central involution w lies in the kernel of every homomorphism to a finite group, so E is a group of *Deligne type* in the terminology of [BDL, Definition 1.2]. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and E is such a counterexample for the operator norm.

Theorem S4.1 (the exact intrinsic obstruction)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, and $\Gamma \leq H$ a subgroup. Then every homomorphism $\pi\colon H \to \mathrm{GL}(V)$ is trivial on the subgroup $\mathfrak{D}(H, \Gamma)$ of* (9)*.*

Proof. Let C be the commutant of $\pi(\Gamma)$. The elements $g \in H$ for which conjugation by $\pi(g)$ preserves C in both directions form a subgroup. The commutant argument of Theorem 3.3 shows that every element of the compression semigroup P_Γ belongs to this subgroup, hence so does all of $G^+(\Gamma) = \langle P_\Gamma \rangle$. Thus, for $g \in G^+(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence every displayed generator in (9) lies in $\ker \pi$; since $\ker \pi \trianglelefteq H$, the whole normal closure $\mathfrak{D}(H, \Gamma)$ does as well. $\square$

Corollary S4.2. *Every finite-dimensional linear representation of the group E of Definition 6.1, over any field, maps w to the identity, and every homomorphism from E to a finite group does as well. In particular E admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space V over any field.*

Proof. Apply Theorem 3.3 with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1; the hypotheses hold by the defining relations (8). Every finite-dimensional representation maps w to the identity, and $w \neq 1$ by Proposition 8.3. For a finite target, compose with its faithful left regular representation over $\mathbb{Q}$. The composite is a finite-dimensional linear representation and hence maps w to the identity; faithfulness of the left regular representation then implies that the original homomorphism also maps w to the identity. $\square$

S5. The cyclic comparison

Theorem S5.1 (the cyclic comparison)**.** *Define the presentation*

$$E_{\mathrm{BS}} = \left\langle \gamma_0, t, c \;\middle|\; \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\mathrm{BS}}^2 = 1, \quad [w_{\mathrm{BS}}, \gamma_0] = [w_{\mathrm{BS}}, t] = [w_{\mathrm{BS}}, c] = 1 \end{array} \right\rangle,$$

where $w_{\mathrm{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$*. Then* E_{BS} *is finitely presented and* $w_{\mathrm{BS}} \neq 1$*. There is an explicit surjective homomorphism from* E_{BS} *onto the subgroup of the Clifford semidirect product generated by the displayed images, and it maps* w_{BS} *to the nontrivial central sign. Nevertheless every finite-dimensional representation of* E_{BS} *over every field maps* w_{BS} *to the identity.*

Proof of Theorem S5.1. The relator $w_{\mathrm{BS}}^2 = 1$ is redundant: tct^{-1} and $\gamma_0(tct^{-1})\gamma_0^{-1}$ are involutions, so centrality forces it by (4). Deleting it gives a six-relator presentation of the same group. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. By the commutant argument of Theorem 3.3, every finite-dimensional representation over every field maps w_{BS} to the identity.

For nontriviality, use the affine realization of $\mathrm{BS}(1, 2)$ and its coset action. The base point is fixed by the cyclic base, while the two points represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these points and map c to the generator at the base point. All seven relators hold, and

$$w_{\mathrm{BS}} \longmapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\mathrm{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjection onto a Clifford quotient. ☐

Sharpness of the Kazhdan hypothesis. Running the Clifford representation of the preceding proof over finite site sets gives explicit unitary models of the seven relators in which w_{BS} is the scalar -1. Fix $m \geq 0$, put $M = 2m + 3$ and $\zeta = e^{2\pi i/M}$, and take the coordinates $\mathbb{Z}/2 \times \mathbb{Z}/M$: the first carries the Clifford sign, the second the spectrum of the base. Let X and Z be the flip and the sign on the first coordinate, put $D = 2^{-1/2}(X + Z)$, a self-adjoint unitary, and set

$$\gamma_0(s, k) = (-1)^s \zeta^k, \qquad t\colon (s, k) \longmapsto (s,\, 2k + b_s), \qquad c = t^{-1}Dt,$$

where γ_0 is the displayed diagonal unitary, $b_0 = 0$, $b_1 = m + 1$, and t is the permutation unitary of the displayed map, which is a permutation because M is odd. Here γ_0 and t are monomial and c is an equal-weight sum of two monomial matrices. Since $tct^{-1} = D$ and $\gamma_0^2(s, k) = \zeta^{2k}$ does not see the spin coordinate, the marked word is

$$w_{\mathrm{BS}} = [\,D,\; ZDZ\,] = (D \cdot ZDZ)^2 = -1,$$

because $ZDZ = 2^{-1/2}(Z - X)$ and $D \cdot ZDZ = \frac{1}{2}(XZ - ZX)$ squares to -1. Being a scalar, w_{BS} satisfies exactly the four relators $w_{\mathrm{BS}}^2 = 1$ and $[w_{\mathrm{BS}}, \gamma_0] = [w_{\mathrm{BS}}, t] = [w_{\mathrm{BS}}, c] = 1$; with $c^2 = 1$ this is five of the seven.

The remaining two hold approximately. From $2(m+1) = M-1$ we get $\zeta^{m+1} = -e^{-i\pi/M}$, and

$$\left\|t\gamma_0 t^{-1} - \gamma_0^2\right\| = \left|1+\zeta^{m+1}\right| = 2\sin\frac{\pi}{2M}, \qquad \|[c,\gamma_0]-1\| \le 2\left|1+\zeta^{m+1}\right|;$$

D commutes with γ_0^2 exactly, so the lamp relator inherits the doubling defect and nothing else. Both are $O(1/m)$. Assembling the models over m gives $\Theta\colon E_{\mathrm{BS}} \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w_{\mathrm{BS}}) \neq 1$, and the image of Θ is MF because it embeds in $\mathcal{U}(\mathcal{Q})$. Thus the property (T) hypothesis in the central involution obstruction is essential: with the cyclic base $\langle\gamma_0\rangle$ in place of a Kazhdan base, the distinguished word survives in a matrix quotient. Equivalently, in the residual language of Section S6, $w_{\mathrm{BS}} \notin \mathrm{Res}_{\mathrm{MF}}(E_{\mathrm{BS}})$.

The realized Clifford quotient is in fact amenable — a subgroup of the semidirect product of the locally finite Clifford group on the coset space by the solvable affine group $\mathrm{BS}(1,2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ — and therefore MF, by quasidiagonality of amenable groups [TWW]. The models above establish only that w_{BS} survives in the MF group $\Theta(E_{\mathrm{BS}})$, and use neither amenability nor quasidiagonality.

The preceding homomorphism to $\mathcal{U}(\mathcal{Q})$ does not arise from exact finite-dimensional representations. In a finite image, the doubling relation gives $\pi(\langle\gamma_0^2\rangle) = \pi(\langle\gamma_0\rangle)$; since $\pi(tct^{-1})$ centralizes the first subgroup, it centralizes $\pi(\gamma_0)$, and the image of w_{BS} is the identity. To transfer this obstruction to exact local embeddings, take a finite set containing the generators together with every intermediate prefix of every defining word and of w_{BS}. Any exact local embedding of this set evaluates those words correctly and therefore satisfies the same equations. Hence neither E_{BS} nor its Clifford quotient is LEF, and the latter is not residually finite.

The scaling family. For $m \ge 2$, define E_m by replacing the three relations $tv_it^{-1} = v_i^2$ in the eight-generator presentation of E with $tv_it^{-1} = v_i^m$ and leaving every other relator unchanged. Then $E_2 = E$. Write w_m for the image of the same distinguished word w.

Corollary S5.2 (the scaling family)**.** *For every natural number $m \ge 2$, the group E_m is finitely presented and w_m is a nontrivial central involution mapped to the identity by every homomorphism $E_m \to \mathcal{U}(\mathcal{Q})$. Consequently E_m is not MF, and neither $C^*_{\max}(E_m)$ nor $C^*_{\mathrm{r}}(E_m)$ is an MF C^*-algebra.*

Proof. Replace the affine dilation $\mathrm{diag}(2,2,2,1)$ by $\mathrm{diag}(m,m,m,1)$ (both acting as in Lemma 8.2). It conjugates each integral translation v_i to v_i^m and fixes the three linear generators. Conjugating v_1 by the inverse dilation gives translation by e_1/m, which is not integral. Hence the two relevant cosets remain distinct, and the Clifford representation sends w_m to -1. The Kazhdan and central-involution arguments are unchanged for every $m \ge 2$. ∎

S6. The MF residual

Definition S6.1. For a countable group G_1, the *MF residual* is

$$\mathrm{Res}_{\mathrm{MF}}(G_1) \;=\; \bigcap_{(d_n)} \;\; \bigcap_{\Theta\colon G_1\to\mathcal{U}(\mathcal{Q}((d_n)))} \ker\Theta,$$

a normal subgroup of G_1; the inner intersection is over all homomorphisms and the outer one over all sequences (d_n) of positive integers, in accordance with the convention fixed after (1).

If G_1/N is MF, then composing $G_1 \to G_1/N$ with an injective homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$ gives a homomorphism with kernel N, so $\mathrm{Res}_{\mathrm{MF}}(G_1) \le N$: the MF residual is contained in every normal subgroup whose quotient is MF, and Proposition S6.3 shows that its own quotient is MF. It is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

Lemma S6.2 (functoriality)**.** *For every homomorphism $\varphi\colon G_1 \to G_2$ of countable groups,*

$$\varphi(\mathrm{Res}_{\mathrm{MF}}(G_1)) \;\subseteq\; \mathrm{Res}_{\mathrm{MF}}(G_2).$$

In particular, if $\varphi\colon E \to G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \mathrm{Res}_{\mathrm{MF}}(G_1)$ and Θ is a homomorphism $G_2 \to \mathcal{U}(\mathcal{Q})$, then $\Theta\circ\varphi$ is a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$, so $\Theta(\varphi(x)) = 1$. For the second statement, Theorem A gives $w \in \mathrm{Res}_{\mathrm{MF}}(E)$. Hence $1 \neq \varphi(w) \in \mathrm{Res}_{\mathrm{MF}}(G_2)$. An injective homomorphism $G_2 \to \mathcal{U}(\mathcal{Q})$ would map $\varphi(w)$ to 1, a contradiction. □

Proposition S6.3 (the universal MF quotient)**.** *Let G_1 be a countable group and $R = \mathrm{Res}_{\mathrm{MF}}(G_1)$.*

(1) R is the kernel of a single homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$.

(2) G_1/R is MF.

(3) Every homomorphism from G_1 to an MF group factors through G_1/R.

In particular G_1 is MF if and only if $R = \{1\}$.

Proof. (3) If $\varphi\colon G_1 \to M$ with M MF and $j\colon M \to \mathcal{U}(\mathcal{Q}((d_n)))$ is an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, then $j\circ\varphi$ is a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$, so $j(\varphi(R)) = 1$ and, by injectivity of j, $\varphi(R) = 1$.

(1) and (2). Set $Q = G_1/R$. For every $\bar g \neq 1$ in Q, the definition of R gives a homomorphism $\Theta_{\bar g}\colon Q \to \mathcal{U}(\mathcal{Q})$ with $\Theta_{\bar g}(\bar g) \neq 1$. Enumerate the nonidentity elements $\bar g_1, \bar g_2, \dots$ and choose coordinate unitary lifts of the corresponding homomorphisms. At stage k, choose in the ith lift, for each $i \le k$, a coordinate at which multiplicativity holds within $1/k$ on the first k group elements and at which the designated element $\bar g_i$ stays separated by its fixed positive constant; such a coordinate exists because $\Theta_{\bar g_i}$ separates $\bar g_i$ along infinitely many coordinates. Take the block sum of the k chosen coordinates. Block operator norms are maxima, so the single block coming from $\Theta_{\bar g_i}$ already separates $\bar g_i$, and the sum therefore separates all of $\bar g_1, \dots, \bar g_k$ at

once. The resulting sequence is an operator-norm asymptotic representation separating every nonidentity element from the identity; its class, composed with the canonical unitary-corona isomorphism $\kappa_{(d_n)}$ of Lemma 2.2, is a faithful homomorphism $Q \to \mathcal{U}(\mathcal{Q})$. This proves (2), and composing with $G_1 \twoheadrightarrow Q$ proves (1). Finally, if G_1 is MF then an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ gives $R = \{1\}$; if $R = \{1\}$ then (2) says G_1 is MF. $\square$

Corollary S6.4 (exact residual from a candidate quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \le \mathrm{Res}_{\mathrm{MF}}(G_1)$. Every homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$ factors uniquely through G_1/N. If in addition G_1/N is MF, then*

$$\mathrm{Res}_{\mathrm{MF}}(G_1) = N.$$

Proof. The containment $N \le \ker \Theta$ gives the unique quotient factorization for every such Θ. For the reverse containment in the displayed equality, compose the quotient map $G_1 \to G_1/N$ with an injective homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$: its kernel is exactly N, while the MF residual lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$. $\square$

Corollary S6.5 (residual reduction to the quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \le \mathrm{Res}_{\mathrm{MF}}(G_1)$ and let $q \colon G_1 \to G_1/N$ be the quotient map. Then*

$$\mathrm{Res}_{\mathrm{MF}}(G_1) \;=\; q^{-1}(\mathrm{Res}_{\mathrm{MF}}(G_1/N)).$$

Proof. For the forward containment, Lemma S6.2 gives $q(\mathrm{Res}_{\mathrm{MF}}(G_1)) \subseteq \mathrm{Res}_{\mathrm{MF}}(G_1/N)$. Conversely, suppose $q(x) \in \mathrm{Res}_{\mathrm{MF}}(G_1/N)$ and let Θ be a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$. Since $N \le \mathrm{Res}_{\mathrm{MF}}(G_1)$, Θ factors as $\bar{\Theta} \circ q$ with $\bar{\Theta}$ a homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$; hence $\Theta(x) = \bar{\Theta}(q(x)) = 1$. $\square$

Corollary S6.6. *Let A be a unital C^*-algebra admitting an injective, possibly nonunital, $*$-homomorphism into a matrix quotient. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.*

Proof. Let $j \colon A \to \mathcal{Q}$ be the injective, possibly nonunital, $*$-homomorphism and let $\phi \colon E \to \mathcal{U}(A)$. Define

$$\widehat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

Then $\widehat{j}$ is an injective homomorphism on $\mathcal{U}(A)$, so $\widehat{j} \circ \phi$ is a homomorphism $E \to \mathcal{U}(\mathcal{Q})$. By Theorem A, $(\widehat{j} \circ \phi)(w) = 1$; injectivity of $\widehat{j}$ gives $\phi(w) = 1$. Proposition 8.3 has $w \neq 1$, so ϕ cannot be injective. $\square$

S6.1. **Permanence and its failure.**

Lemma S6.7 (permanence)**.** *Let all groups be countable.*

1. *Every subgroup of an MF group is MF.*
2. *Every residually finite group is MF.*
3. *Every locally finite group is MF.*

Proof. (1) Restrict an MF embedding of the ambient group to the subgroup. (2) For a finite test set F, a finite quotient injective on F, composed with its left regular permutation action, is an exact finite-dimensional representation; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-global direction of Proposition 2.3 concludes. (3) A finite test set F lies in a finite subgroup K. The left regular representation $\lambda_K\colon K \to \mathcal{U}(\mathbb{C}^K)$ is a finite-dimensional unitary representation of K by permutation matrices. Because a local model is defined on all of G, set

$$V(g) = \lambda_K(g) \quad (g \in K), \qquad V(g) = 1 \quad (g \notin K).$$

On the test set $F \subseteq K$, this map is exactly multiplicative, since products of test elements lie in K and λ_K is a homomorphism; the values outside K do not enter the two conditions. It separates as in (2). □

Corollary S6.8 (quotients)**.** *MF groups are not closed under quotients: the free group F_8 is MF, and the explicit group E is its quotient but is not MF.*

Proof. Map the eight free generators onto the eight displayed generators of E. The resulting homomorphism is surjective because those generators define the displayed presentation. The group F_8 is residually finite, hence MF by Lemma S6.7(2), while E is not MF by Theorem A. □

S7. Commuting orbits and the MF residual

In Theorem 4.2 the corner is the negative spectral corner of a central involution. Here it is an arbitrary projection of $\mathcal{Q}$ whose conjugation orbit commutes.

Theorem S7.1 (a commuting orbit forces commutation)**.** *Let H be countable, let $L \leq H$ have property* (T)*, let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Let $p \in \mathcal{Q}$ be a projection such that p commutes with $\Theta(s\gamma s^{-1})$ for every $\gamma \in L$ and the projections $\Theta(\gamma)\, p\, \Theta(\gamma)^*$, $\gamma \in L$, commute pairwise. Then p commutes with $\Theta(L)$.*

Proof. Property (T) makes L finitely generated; fix a finite symmetric generating set $S = \{a_1, \dots, a_m\}$. Write $p_g = \Theta(g)\, p\, \Theta(g)^*$ and $\Delta_g = p_g - p$ for $g \in L$, and suppose $\Delta_{g_0} \neq 0$ for some $g_0 \in L$.

Since p_g and p are commuting projections, Δ_g is self-adjoint with $\Delta_g^3 = \Delta_g$, so Δ_g^2 is a projection and $\Delta_g \Delta_g^2 = \Delta_g$; write $q_g = \Delta_g^2$. All the p_g commute, hence so do all the Δ_g and all the q_g. Put

$$q = 1 - \prod_{a \in S} (1 - q_a),$$

the join of the projections q_a, $a \in S$. The elements g with $p_g = p$ form a subgroup of L, so if $q = 0$, and hence $\Delta_a = 0$ for every $a \in S$, then $\Delta_g = 0$ for every $g \in L$, contradicting $\Delta_{g_0} \neq 0$. Thus $q \neq 0$.

The q-normalized Hilbert space. Take a self-adjoint bounded lift (x_n) of q. Since $\|x_n^2 - x_n\| \to 0$, for all sufficiently large n the spectrum of x_n avoids $[1/4, 3/4]$. Choose a continuous function $f\colon \mathbb{R} \to [0,1]$ with $f = 0$ on $(-\infty, 1/4]$ and $f = 1$ on $[3/4, \infty)$. Continuous functional calculus then gives, for all sufficiently large n, a projection $Q_n = f(x_n)$ with $\|Q_n - x_n\| \to 0$; define $Q_n = 0$ on the finitely many remaining coordinates. Thus (Q_n) is a projection lift of q.

Let $J = \{n : Q_n \neq 0\}$. Because $q \neq 0$, the set J is infinite. Restriction of coordinates induces a $*$-homomorphism

$$\rho_J\colon \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}) \longrightarrow \prod_{n \in J} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in J} M_{d_n}(\mathbb{C}).$$

Replace Θ by $\rho_J \circ \Theta$ and every element introduced above by its image under ρ_J. All projection, commutation and cocycle identities are preserved, while the new image of q is still nonzero because its lift consists of nonzero projections and therefore has norm 1 at every coordinate of J. Relabel J by $\mathbb{N}$; then, without changing notation, we have $k_n = \operatorname{rank} Q_n \geq 1$ for every n.

Give $M_{d_n}(\mathbb{C})$ the inner product $\langle x, y\rangle_n = \operatorname{Tr}(y^*x)/k_n$, write $\|x\|_{\mathrm{F}} = \operatorname{Tr}(x^*x)^{1/2}$ for the unnormalized Hilbert–Schmidt norm, and let K_ω be the Hilbert-space ultraproduct of these spaces along a free ultrafilter ω.

The corner-supported ideal. Let $\mathcal{I}_q$ be the algebraic two-sided ideal generated by q, namely the set of finite sums $\sum_{j \leq r} x_j q y_j$ with $x_j, y_j \in \mathcal{Q}$. It is invariant under inner automorphisms. It is not closed in norm; the finite-sum description gives the rank bound (11) below.

For $z = \sum_{j \leq r} x_j q y_j \in \mathcal{I}_q$, choose bounded coordinate lifts $X_{j,n}$ of x_j and $Y_{j,n}$ of y_j and put $Z_n = \sum_{j \leq r} X_{j,n} Q_n Y_{j,n}$; then $\operatorname{rank}(Z_n) \leq r k_n$. The square of the Frobenius norm is the sum of the squares of the singular values; at most $\operatorname{rank}(Z_n)$ of them are nonzero and each is at most the operator norm, so

$$\|Z_n\|_{\mathrm{F}}^2 \;\leq\; \operatorname{rank}(Z_n)\, \|Z_n\|^2 \;\leq\; r\, k_n\, \|Z_n\|^2 \tag{11}$$

and (Z_n) is a bounded sequence for the inner products $\langle \cdot, \cdot \rangle_n$. Its class in K_ω depends only on $z = \sum_j x_j q y_j$: a sequence Z'_n obtained from a second decomposition, with r' summands, satisfies $\|Z_n - Z'_n\| \to 0$ and $\operatorname{rank}(Z_n - Z'_n) \leq (r + r')k_n$, so $\|Z_n - Z'_n\|_{\mathrm{F}}^2 / k_n \to 0$ by (11). Write $\Lambda(z) = [Z_n]_\omega$. The map $\Lambda\colon \mathcal{I}_q \to K_\omega$ is linear,

$$\|\Lambda(q)\| = \lim_\omega \Big(\frac{\operatorname{Tr} Q_n}{k_n}\Big)^{1/2} = 1,$$

and $\Lambda(z) = 0$ implies $\Lambda(xzy) = 0$ for all $x, y \in \mathcal{Q}$, since bounded lifts of x and y multiply $\|Z_n\|_{\mathrm{F}}$ by a bounded factor.

An exact cocycle. For $a \in S$ the projection q_a annihilates $\prod_{b \in S}(1 - q_b)$, so $q_a q = q_a$ and $\Delta_a = \Delta_a q_a = \Delta_a q \in \mathcal{I}_q$. From $p_{gh} = \Theta(g)\, p_h\, \Theta(g)^*$,

$$\Delta_{gh} = \Delta_g + \Theta(g)\, \Delta_h\, \Theta(g)^* \qquad (g, h \in L), \tag{12}$$

and $\mathcal{I}_q$ is inner-invariant, so $\Delta_g \in \mathcal{I}_q$ for every $g \in L$.

Choose unitary coordinate lifts $U_n(h)$ of $\Theta(h)$ (Lemma 2.2). Conjugation by a unitary preserves each $\langle\cdot,\cdot\rangle_n$, and $\|\mathrm{Ad}\, u - \mathrm{Ad}\, v\| \le 2\,\|u-v\|$, so $\pi(h) = [\mathrm{Ad}\, U_n(h)]_\omega$ defines a unitary representation of H on K_ω, and $\pi(h)\Lambda(z) = \Lambda(\Theta(h)\, z\, \Theta(h)^*)$ on the closed invariant subspace $K_q = \overline{\Lambda(\mathcal{I}_q)}$. By (12) the map $\beta(g) = \Lambda(\Delta_g)$ satisfies

$$\beta(gh) = \beta(g) + \pi(g)\beta(h) \qquad (g, h \in L),$$

a 1-cocycle for $\pi|_L$ with values in K_q.

The cocycle is nonzero. Suppose $\beta(a) = 0$ for every $a \in S$. Put $r_1 = 1$ and $r_i = \prod_{j<i}(1 - q_{a_j})$ for $i \ge 2$. The products $e_i = r_i q_{a_i}$ are projections with $e_i = r_i - r_{i+1}$, so they are pairwise orthogonal and $\sum_i e_i = 1 - r_{m+1} = q$. Each Δ_{a_i} commutes with r_i, so $e_i = (\Delta_{a_i} r_i)\,\Delta_{a_i}$ and $\Lambda(e_i) = 0$, whence $\Lambda(q) = 0$ against $\|\Lambda(q)\| = 1$. So β is not identically zero.

Property (T). By the Delorme–Guichardet theorem [BHV, Theorem 2.12.4], β is a coboundary: there is $y \in K_q$ with $\beta(g) = y - \pi(g)y$ for every $g \in L$. The hypothesis on p gives $\Delta_{s\gamma s^{-1}} = 0$, hence $\pi(s\gamma s^{-1})y = y$ for every $\gamma \in L$.

Normalize the Hilbert–Schmidt norms by $k_n = \mathrm{rank}\, Q_n$ rather than by d_n and apply Theorem S2.1 with that weight. In its proof, P is the projection onto $\mathrm{Fix}\, \pi(L)$ and the conjugate $Q = \pi(s)P\pi(s)^*$ onto $\mathrm{Fix}\, \pi(sLs^{-1})$, with $P \le Q$ and $P \sim Q$ in the same finite norm ultraproduct. Lemma 3.1 therefore gives $P = Q$. Hence $\pi(g)y = y$ and $\beta(g) = 0$ for every $g \in L$, contradicting the previous paragraph. Therefore $\Delta_g = 0$ for every $g \in L$ in the original corona; equivalently, p commutes with $\Theta(g)$ for every $g \in L$. $\square$

Equivalently, every projection of $\Theta(sLs^{-1})' \cap \mathcal{Q}$ whose $\Theta(L)$-conjugates commute pairwise already lies in $\Theta(L)' \cap \mathcal{Q}$.

Definition S7.2 (involutive orbit witness)**.** Let H be a group, $L \le H$ a subgroup, and $s \in H$. An element $k \in H$ is an *involutive orbit witness* for (L, s) if

(W1) $k^2 = 1$;
(W2) $s\gamma s^{-1}$ commutes with k for every $\gamma \in L$;
(W3) the conjugates $\gamma_1 k \gamma_1^{-1}$ and $\gamma_2 k \gamma_2^{-1}$ commute for all $\gamma_1, \gamma_2 \in L$.

The pair (L, s) determines the normal subgroup

$$D_{\mathrm{coll}}(L, s) \;=\; \big\langle\!\big\langle\, [\gamma, k] \;:\; \gamma \in L,\; k \text{ a witness for } (L, s) \,\big\rangle\!\big\rangle_H.$$

In a permutational wreath product $K^{(X)} \rtimes G$ with $L \le G$ conjugated into itself by s and $x_0 \in X$ a point fixed by sLs^{-1}, every involution in the copy of K supported at x_0 is a witness: distinct coordinate copies commute, and sLs^{-1} fixes x_0.

Theorem S7.3 (the orbit of an involutive witness)**.** *Let H be countable, let $L \le H$ be a subgroup with property* (T)*, and let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$. Then*

$$D_{\mathrm{coll}}(L, s) \;\le\; \mathrm{Res}_{\mathrm{MF}}(H).$$

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and k a witness for (L, s). By (W1) the unitary $u = \Theta(k)$ is self-adjoint, so $p = \frac{1}{2}(1-u)$ is a projection, and

$$\Theta(\gamma)\, p\, \Theta(\gamma)^* = \tfrac{1}{2}\big(1 - \Theta(\gamma k \gamma^{-1})\big) \qquad (\gamma \in L).$$

By (W2) the projection p commutes with $\Theta(s\gamma s^{-1})$ for $\gamma \in L$, and by (W3) the displayed projections commute pairwise. By Theorem S7.1, p commutes with $\Theta(L)$; hence so does $u = 1 - 2p$, and $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. Kernels are normal, so Θ is trivial on $D_{\mathrm{coll}}(L, s)$. ☐

Remark S7.4 (finite-order witnesses). Condition (W1) can be relaxed to finite order. Let H be countable, let $L \leq H$ have property (T), let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $k \in H$ have finite order m. If sLs^{-1} centralizes k and the conjugates $\gamma k \gamma^{-1}$, $\gamma \in L$, commute pairwise, then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. For $j \in \mathbf{Z}/m$, let

$$p_j = \frac{1}{m} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t/m} \Theta(k)^t.$$

These are the spectral projections of $\Theta(k)$. The hypotheses imply that $\Theta(sLs^{-1})$ centralizes every p_j and that their $\Theta(L)$-conjugates commute pairwise. Theorem S7.1 therefore makes every p_j commute with $\Theta(L)$. Fourier reconstruction, $\Theta(k) = \sum_j e^{2\pi i j/m} p_j$, then shows that $\Theta(k)$ also commutes with $\Theta(L)$. The normal closure of all such commutators, over all finite-order witnesses for (L, s), lies in $\mathrm{Res}_{\mathrm{MF}}(H)$, with equality whenever the corresponding quotient of H is MF.

Proof of Theorem B. Write $q\colon E \to E/\langle w \rangle$, $\bar{c} = q(\hat{c})$, and $\bar{L} = q(\iota(\mathcal{B}))$. The element $\bar{c}$ is an involution because $c^2 = 1$, and $q(t)\bar{L}q(t)^{-1}$ centralizes $\bar{c}$ because conjugating the relation $[c, \iota(\mathcal{B})] = 1$ of (8) by t gives $[\, t\iota(\mathcal{B})t^{-1},\, \hat{c}\,] = 1$ in E, where $\hat{c} = tct^{-1}$. The subgroup that centralizes $\hat{c}$ is the conjugate one, which is why the relation is conjugated by t. Property (T) passes to $\bar{L}$, and $q(t)$ conjugates it into itself. Thus (W1) and (W2) of Definition S7.2 hold, and only (W3) remains.

Pairwise commutation of the orbit. The linear generators x, y, z commute with $\hat{c}$, and conjugation by t sends each v_i to v_i^2, so $\iota(v_i^2)$ commutes with $\hat{c}$ as well. The conjugate $\iota(g)\, \hat{c}\, \iota(g)^{-1}$ therefore depends only on the exponent vector of the translation part of g modulo 2, so there are at most eight such conjugates, indexed by $(\mathbb{Z}/2)^3$. By Theorem 5.1 the distinguished word is $w = [\hat{c}, \iota(v_1)\, \hat{c}\, \iota(v_1)^{-1}]$, the commutator attached to the class of v_1. The linear part acts through $\mathrm{SL}_3(\mathbb{Z}/2)$, which is transitive on the nonzero parity classes; the six words x, x^2, z, yz, yx, y^2x in the linear generators send the class of v_1 to the six remaining nonzero classes, and each commutes with $\hat{c}$. Conjugating the displayed identity by the corresponding $\iota(r)$ shows that every nonzero parity commutator is a conjugate of w; the zero class gives $[\hat{c}, \hat{c}] = 1$. Since w is central, all eight lie in $\langle w \rangle$. Conjugating the commutator of two arbitrary conjugates by the inverse of one of them reduces it to this case. Hence all conjugates of $\bar{c}$ under $\bar{L}$ commute pairwise. Thus $\bar{c}$ is an involutive orbit witness for $(\bar{L}, q(t))$, and by Theorem S7.3 $[q(\iota(v_1)), \bar{c}] \in \mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle)$.

The homomorphism $E \to W$ of Proposition 8.3 sends $[\iota(v_1), \hat{c}]$ to the product of the Clifford generators at the two distinct points τo and $j(\bar{v}_1)\tau o$, and sends $\langle w\rangle$ into $\langle \zeta\rangle$. The product of the Clifford generators at two distinct points does not lie in $\langle\zeta\rangle$, so $[\iota(v_1), \hat{c}] \notin \langle w\rangle$ and its image in $E/\langle w\rangle$ is nontrivial. By Proposition S6.3, this nontrivial element of the MF residual makes the quotient non-MF. Corollary S6.5 identifies $\mathrm{Res}_{\mathrm{MF}}(E)$ with the q-preimage of $\mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle)$, so u belongs to it. $\square$

Corollary S7.5 (collapse computes the residual)**.** *In the setting of Theorem S7.3, write $q\colon H \to H/D_{\mathrm{coll}}(L,s)$ for the quotient map. Then*

$$\mathrm{Res}_{\mathrm{MF}}(H) \;=\; q^{-1}\big(\mathrm{Res}_{\mathrm{MF}}(H/D_{\mathrm{coll}}(L,s))\big),$$

and if $H/D_{\mathrm{coll}}(L,s)$ is MF, then $\mathrm{Res}_{\mathrm{MF}}(H) = D_{\mathrm{coll}}(L,s)$.

Proof. Combine Theorem S7.3 with Corollaries S6.5 and S6.4. $\square$

If $H/D_{\mathrm{coll}}(L,s)$ is MF, then it is the universal MF quotient of H (Proposition S6.3): every homomorphism from H to an MF group factors through it.

S8. Group algebras, soficity, and permanence

S8.1. The reduced group algebra of E.

Lemma S8.1. *Let G_1 be a countable discrete group.*

(1) The canonical tracial state τ on $C^*_{\mathrm{r}}(G_1)$ is faithful.
(2) Every unital C^*-algebra A with a faithful tracial state is stably finite: for every $k \geq 1$, every isometry in $M_k(A)$ is a unitary.

Proof. (1) Realize $C^*_{\mathrm{r}}(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g. Let ρ_g be the right translation unitaries, $(\rho_g\xi)(h) = \xi(hg)$. Each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C^*_{\mathrm{r}}(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e\rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k}\sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k}\sum_{i,j}\tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i-th diagonal entry of x^*x is $\sum_j x^*_{ji}x_{ji}$, so $\tau_k(x^*x) = 0$ implies $\tau(x^*_{ji}x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j. If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

S8.2. **The Clifford quotient and permanence.** The map $E \to W$ of Proposition 8.3 is surjective: its image contains Σ and the generator c_o at the base point, hence every Clifford generator by transitivity of the Σ-action on X. Since E is generated by v_1, x, y, z, t, c (Section 7), their six images generate W. The map sends $w \in \mathrm{Res}_{\mathrm{MF}}(E)$ to the nontrivial sign ζ, so $\zeta \in \mathrm{Res}_{\mathrm{MF}}(W)$ by Lemma S6.2. Hence the finitely generated group W is not MF. Neither is $C^*_{\mathrm{r}}(W)$: if $C^*_{\mathrm{r}}(W)$ embedded in a matrix quotient, the map $u \mapsto e(u) + 1 - e(1)$ would give an injective homomorphism $W \to \mathcal{U}(\mathcal{Q})$, as in the proof of Theorem A.

Failure of extension permanence. MF groups are not closed under extensions. More precisely, W fits into an exact sequence

$$1 \longrightarrow \mathrm{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel is locally finite — hence LEF, sofic and MF — and whose quotient is MF, while W itself is finitely generated and not MF. For comparison, an extension of a residually finite group by a residually finite group is weakly sofic [Gl], and weak soficity is detected by finite targets [GR].

Proof. The Clifford group $\mathrm{Cl}(X)$ is locally finite: every finitely generated subgroup involves only finitely many sites, hence is contained in the corresponding finite Clifford group. Therefore $\mathrm{Cl}(X)$ is MF by Lemma S6.7(3). The group Σ identifies with the subgroup of $\mathrm{GL}_4(\mathbb{Q})$ generated by $\bar{\Gamma}$ and D: the levels are $D^{-n}\bar{\Gamma}D^n$, and the determinant gives the $\mathbb{Z}$-coordinate. It is residually finite. Modulo every odd integer, the dilation matrix is invertible because its determinant is 8; hence the congruence reduction of the integral affine model is compatible with the dilation and extends over the direct limit and its shift. These finite quotients separate every nontrivial element of Σ, consistent with Mal'cev's theorem for finitely generated linear groups [Mal]. Thus Σ is MF by Lemma S6.7(2), whereas W is not MF by the preceding construction. □

Extensions by $\mathbb{Z}$. Composing the two canonical projections gives a surjection $W \to \mathbb{Z}$ whose kernel is $\mathrm{Cl}(X) \rtimes T$, and this kernel is LEF, hence MF: every finite subset lies in $L \rtimes \bar{\Gamma}_n$ for a finite invariant subgroup $L \leq \mathrm{Cl}(X)$ and one level, and that group is residually finite because the level acts on the finite group L through a finite quotient. Thus W is an extension of an LEF group by $\mathbb{Z}$ and is not MF, while soficity is preserved by every extension with amenable quotient, as in Elek–Szabó [ES06].

A simple sofic envelope. There is a countable simple sofic group S_{simp} with

$$\mathrm{Res}_{\mathrm{MF}}(S_{\mathrm{simp}}) = S_{\mathrm{simp}}.$$

Equivalently, every homomorphism from S_{simp} to an MF group is trivial.

Proof. By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed E, which is sofic by Theorem D, in such a group. By functoriality of the MF residual, the image of w is a nontrivial element of the MF residual of the simple group. This residual is a nontrivial normal subgroup, so simplicity makes it the whole group. Every homomorphism to an MF group maps the MF residual to 1. ∎

S8.3. **Soficity of E.** Deleting the letter c and the families (C) and (W) from (8) leaves the presentation $\langle \mathcal{B}, t \mid tut^{-1} = \bar\alpha(u) \rangle$ of the ascending HNN extension of the base along the doubling endomorphism. By Remark 7.2 the base is $\bar\Gamma$, so

$$E/\langle\!\langle c \rangle\!\rangle \;\cong\; \Sigma$$

for the group $\Sigma = T \rtimes \mathbb{Z}$ of Section 8, with its site set $X = \Sigma/B$, base point o, shift τ and level-zero base $B = j(\bar\Gamma)$. The levels

$$\bar\Gamma_n = \tau^{-n} B \tau^n \qquad (n \geq 0)$$

form an ascending chain of copies of $\bar\Gamma$ with union T. Put $K = \tau^{-1} B \tau$; then $B \leq K$ and $[K : B] = [\bar\Gamma : \bar\alpha(\bar\Gamma)]$, at most eight. The cosets Σ/K are the *blocks*, and each block, as a set of sites, has $[K : B]$ elements.

Let $\mathcal{G}$ be the graph on X whose edges are the Σ-orbit of the pair $\{\tau o,\, j(\bar v_1)\tau o\}$, whose entries are distinct by Lemma 8.2. Two sites are adjacent in $\mathcal{G}$ exactly when they are distinct and lie in a common block. Adjacent sites share a block because the two sites are gB and hB for the representatives $g = \tau$ and $h = j(\bar v_1)\tau$, and $g^{-1}h = \tau^{-1} j(\bar v_1)\tau \in K$ since $j(\bar v_1) \in B$ and $K = \tau^{-1} B \tau$; hence $gK = hK$. (Sites are cosets, so the product and inverse have to be taken in Σ on representatives, not on the sites themselves.) Conversely, the bijection $u\,\bar\alpha(\bar\Gamma) \mapsto \tau^{-1} j(u) \tau o$ identifies the block of o with $\bar\Gamma/\bar\alpha(\bar\Gamma)$ and the generating edge with the pair $\{\bar\alpha(\bar\Gamma),\, \bar v_1 \bar\alpha(\bar\Gamma)\}$. Left multiplication by $\bar\alpha(\bar\Gamma)$ on the coset space $\bar\Gamma/\bar\alpha(\bar\Gamma)$ is transitive on the nontrivial cosets, and that is what makes every pair of distinct sites of a block the image of the generating pair. Writing $L = \langle x, y, z \rangle$ for the linear part, $\bar\Gamma = \mathbb{Z}^3 \rtimes L$; since $\bar\alpha$ squares the three translations and fixes x, y, z (Lemma 8.2), its image is $(2\mathbb{Z}^3) \rtimes L$, and reduction of the translation coordinate modulo 2 identifies $\bar\Gamma/\bar\alpha(\bar\Gamma)$ with $(\mathbb{Z}/2)^3$. Under that identification left multiplication by an element of $\bar\alpha(\bar\Gamma)$ with linear part N and (necessarily even) translation part sends a class $\bar u$ to $\bar N \bar u$, so the action factors through the linear action of the reduction $\bar L \leq \mathrm{GL}_3(\mathbb{F}_2)$ of L. The seven words

$$1, \quad x, \quad x^2, \quad z, \quad yx, \quad y^2x, \quad xyx$$

carry $\bar e_1$ to $\bar e_1$, $\bar e_3$, $\bar e_2$, $(0,1,1)$, $(1,1,0)$, $(1,1,1)$, $(1,0,1)$ — the seven nonzero classes — which is the transitivity used. Only the reduction $\bar L$ enters, so this step is independent of the identification of the linear part with $\mathrm{SL}_3(\mathbb{Z})$ in Remark 7.2.

For a graph $\mathcal{G}$ on X, let $C(\mathcal{G})$ be the group presented by generators ζ and c_ξ $(\xi \in X)$ and relations

$$\zeta^2 = c_\xi^2 = 1, \qquad [\zeta, c_\xi] = 1, \qquad [c_\xi, c_\eta] = \zeta \;\text{ for each edge } \{\xi, \eta\} \text{ of } \mathcal{G}.$$

For the complete graph on X this is $\mathrm{Cl}(X)$ of Construction 8.1, and adjoining the anticommutation relations at the remaining pairs gives a surjection $C(\mathcal{G}) \to \mathrm{Cl}(X)$.

Proposition S8.2 (normal form for E)**.** *There is an isomorphism*

$$E \;\cong\; C(\mathcal{G}) \rtimes \Sigma$$

under which $\langle\langle c\rangle\rangle$ is the factor $C(\mathcal{G})$, the conjugate gcg^{-1} is the lamp c_{gB}, and the distinguished word w is the sign ζ. Each level $\bar{\Gamma}_n$ has finite orbits on X.

Proof. Send the six base letters and t to the corresponding generators of Σ, and c to c_o. Families (B) and (T) of (8) hold in Σ by construction. For family (C): $c_o^2 = 1$, and B fixes o, so each base generator fixes c_o. For family (W): the pair $\{\tau o,\, j(\bar{v}_1)\tau o\}$ is an edge, so the image of w is $[c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = \zeta$, which is central in $C(\mathcal{G})$ and fixed by Σ. This defines $E \to C(\mathcal{G}) \rtimes \Sigma$.

In the other direction, $\zeta \mapsto w$ and $c_{gB} \mapsto gcg^{-1}$ respects the defining relations of $C(\mathcal{G})$: the assignment is independent of the coset representative because the base centralizes c; the images are involutions; w is central of order two; and along an edge $g\{\tau o,\, j(\bar{v}_1)\tau o\}$ the commutator of the two images is $gwg^{-1} = w$; the presentation imposes relations only along edges. Combined with the splitting $\Sigma \to E$ on the seven remaining letters, this gives $C(\mathcal{G}) \rtimes \Sigma \to E$. The two composites fix the eight letters of E and the generators ζ, c_ξ of $C(\mathcal{G})$, hence are the identity.

For the orbits, $\bar{\alpha}(\bar{\Gamma})$ has index at most eight in $\bar{\Gamma}$ — the parity classes of the translation lattice — so τ conjugates B onto a subgroup of finite index in itself. Every $\bar{\Gamma}_n$ is therefore commensurable with B, all of Σ commensurates B, and orbit–stabilizer makes each $\bar{\Gamma}_n$-orbit in $X = \Sigma/B$ finite. □

Composing the isomorphism with $C(\mathcal{G}) \to \mathrm{Cl}(X)$ gives the surjection $E \to W$ of Proposition 8.3. The group $\mathrm{Cl}(X)$ is locally finite; $C(\mathcal{G})$ is not, since two lamps at sites of different blocks generate an infinite dihedral group and three blocks give a nonabelian free subgroup. For a finite set S of sites write

$$M_S = \langle\, \zeta,\; c_\xi \;(\xi \in S)\,\rangle \;\le\; C(\mathcal{G}).$$

Lemma S8.3 (finite-site subgroups are residually finite)**.** *For every finite set S of sites the subgroup M_S is residually finite.*

Proof. The quotient map by $\langle\zeta\rangle$ maps $C(\mathcal{G})$ onto the free product of the elementary abelian 2-groups of the blocks, and M_S into the free product of the finitely many factors at blocks meeting S. A free product of finite groups is residually finite. The kernel of $C(\mathcal{G}) \to C(\mathcal{G})/\langle\langle\zeta\rangle\rangle$ is the normal closure of ζ, and ζ is central, so that normal closure is $\langle\zeta\rangle$; hence the kernel of this map on M_S lies in $\langle\zeta\rangle$. The surjection $C(\mathcal{G}) \to \mathrm{Cl}(X)$ sends ζ to the sign of $\mathrm{Cl}(X)$, which is nontrivial, and $\mathrm{Cl}(X)$ is locally finite, so the image of the finitely generated subgroup M_S in $\mathrm{Cl}(X)$ is finite. An element of M_S in the kernel of the first map lies in $\langle\zeta\rangle$, and the second map separates the two elements of that subgroup. □

Proof of Theorem D. Let E_0 be the kernel of the homomorphism $E \to \mathbb{Z}$ given by the exponent sum in t. By Proposition S8.2, $E_0 \cong C(\mathcal{G}) \rtimes T$.

Take a finite subset of E_0. Its T-coordinates lie in one level $\bar{\Gamma}_n$, the chain being ascending, and its $C(\mathcal{G})$-coordinates involve only finitely many lamp generators and possibly the sign, so they lie in M_{S_0} for some finite set S_0 of sites. Let S be the union of the $\bar{\Gamma}_n$-orbits of the sites in S_0, a finite set by Proposition S8.2. Then M_S is $\bar{\Gamma}_n$-invariant and contains M_{S_0}, so the finite subset lies in $M_S \rtimes \bar{\Gamma}_n$.

That group is residually finite. Indeed M_S is residually finite by Lemma S8.3, the level $\bar{\Gamma}_n \cong \bar{\Gamma}$ is residually finite through the congruence quotients of its matrix model (Lemma 8.2), and the action of $\bar{\Gamma}_n$ on M_S factors through a finite permutation group F because S is finite. The kernel of $\bar{\Gamma}_n \to F$ has finite index, and the corresponding finite-index subgroup of the semidirect product is $M_S \times \ker(\bar{\Gamma}_n \to F)$, which is residually finite; residual finiteness passes to a group containing a residually finite subgroup of finite index.

Hence E_0 is a directed union of subgroups that embed in residually finite groups; that is, E_0 is LEF and therefore sofic. The group E is a split extension of E_0 by $\mathbb{Z}$, split by t, so the Elek–Szabó amenable-extension theorem implies that E is sofic [ES06], and hence hyperlinear. By Theorem A, E is not MF. LEF groups are MF [CDE], so E is not LEF; a residually finite group is LEF, so E is not residually finite either. □

Exactness. The same decomposition also proves that E is exact. The quotient $C(\mathcal{G})/\langle\zeta\rangle$ is the free product of the elementary abelian 2-groups attached to the blocks, each of order at most 2^8. There are countably many blocks; enumerate them, and let Q_n be the free product of the factors attached to the first n. Each factor is finite, hence exact, so each Q_n is a free product of finitely many exact groups and is exact by Dykema's theorem [Dy]. The Q_n increase with union the whole quotient, and a group with an increasing family of exact open subgroups whose union it is is itself exact [KW1, Lemma 2.5] — in a discrete group every subgroup is open — so this quotient is exact. Since $\langle\zeta\rangle$ is finite, extension permanence [KW2] then gives exactness of $C(\mathcal{G})$. The group $\Sigma \leq \mathrm{GL}_4(\mathbb{Q})$ is exact by Guentner–Higson–Weinberger [GHW]. Finally $E \cong C(\mathcal{G}) \rtimes \Sigma$ is an extension, hence is exact by [KW2]. For a discrete group, group exactness is equivalent to exactness of its reduced group C^*-algebra [KW1]; therefore $C^*_{\mathrm{r}}(E)$ is exact.

S9. Marked limits, quasi-identities, and undecidability

S9.1. **Marked-limit closure.**

Theorem S9.1 (the MF locus is closed)**.** *Fix $k \geq 1$ and consider the space of k-marked groups — quotients of the free group F_k, with the usual topology in which a basic clopen set specifies the truth values of finitely many word equations in the generators.*

(1) The marked groups with MF quotient form a closed set.
(2) Equivalently, the non-MF marked groups form an open set.
(3) Every non-MF k-marked group has a finite-radius obstruction: there is a radius R such that every k-marked group whose relations agree with it on the reduced words of length at most R is non-MF.

Proof. By Proposition 2.3, a marked group fails MF exactly when the local separation formulation fails for some finite subset and some positive tolerance. Choose words representing the finitely many elements of that subset and all products that occur in the corresponding multiplicativity conditions. Whether those words represent the same group elements depends on only finitely many word equalities. Hence the same failed local approximation persists throughout a basic clopen neighborhood of the marked group. Taking R at least the lengths of all words involved gives the finite-radius statement. □

S9.2. **A finite universal Horn obstruction.** The presentation of E yields a finite quasi-identity — equivalently, a universal Horn sentence — satisfied by every MF group. Write R for the forty-one relators of (8), regarded as words in eight variables, and w for the distinguished word in those variables.

Proposition S9.2 (finite quasi-identity)**.** *Every MF group G satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8 \colon \quad \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

while the canonical generating tuple of E satisfies every relator in R and has $w \neq 1$.

Proof. A tuple satisfying the relators induces a homomorphism $E \to G$ sending the distinguished word of E to $w(x_1, \dots, x_8)$. If G is MF, the image is trivial, because $w \in \mathrm{Res}_{\mathrm{MF}}(E)$ by Theorem A and the residual is functorial (Lemma S6.2). The canonical tuple of E satisfies R by definition, and its distinguished word is nontrivial by Theorem A. □

In the space of eight-generator marked groups, the groups satisfying all relators of R and $w \neq 1$ therefore form a nonempty clopen set consisting entirely of non-MF groups: both conditions depend on finitely many word equalities and inequalities, the set contains the marked group E, and Proposition S9.2 excludes every MF point.

S9.3. **Undecidability of MF recognition.** Lemma S6.7(1) and Theorem A show that MF is a Markov property of finitely presented groups: the trivial group is MF, while by subgroup heredity the finitely presented group E

embeds in no MF group. The Adian–Rabin theorem therefore implies that MF recognition is undecidable [Ra].

Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_e for the group presented by the code e.

Corollary S9.3 (undecidability of MF recognition)**.** *Write W for the predicate on pairs consisting of a presentation code and a word in the generators of that code, holding when the word is trivial in the presented group. Then:*

(1) W is undecidable;
(2) no algorithm decides, from a presentation code, whether the presented group is MF;
(3) W is recursively enumerable, whereas its complement and the set of presentation codes of non-MF groups are not.

Proof. For the undecidability of W, take an Aanderaa–Cohen modular machine whose configuration halting problem is undecidable and the finitely presented group built from it [AC]; that paper is where modular machines are introduced and where the halting problem of such a machine is converted into the word problem of a finitely presented group. That group has a presentation code; the halting word of a configuration is a computable function of the configuration in the numbering that code uses; and the code's word problem holds on that word exactly when the configuration halts. A decision procedure for W would compose with that map to decide halting.

For the reduction we use the effective form of the Adian–Rabin construction, which supplies a computable map Δ from pairs (a presentation code e, a word u in its generators) to presentation codes such that

- if $u = 1$ in G_e, then $\Delta(e, u)$ presents an MF group: the trivial group in the classical form, and a free group in the variant used here, a free group being residually finite and so MF by Lemma S6.7(2); and
- if $u \neq 1$ in G_e, then G_e embeds in the group presented by $\Delta(e, u)$.

This is Rabin's theorem [Ra] and, in the form with the explicit computable output presentation, Adian's [Ad]; an English translation of Adian's papers with the construction written out is [NB]. Apply Δ to the free product of the input presentation with a fixed presentation of E and to the input word. The resulting presentation code is computable from the pair and presents an MF group exactly when the input word is trivial: in the trivial case the output group is free, hence MF, and in the nontrivial case the free product — and so E — embeds in the output group, which is therefore non-MF by Lemma S6.7(1). Composing a decision procedure for MF recognition with this map would therefore decide W, so no such procedure exists. For the last assertion, the positive instances of W are exactly those carrying a finite certificate — a product of conjugates of relators freely equal to the given word — and whether a given certificate works is decidable, so W is recursively enumerable. Were its complement recursively enumerable as well,

W would be decidable, which it is not. Finally, the preimage of the set of non-MF presentation codes under this computable map is exactly the set of pairs outside W, and the preimage of a recursively enumerable set under a computable map is recursively enumerable. □

Multiplicity. There are infinitely many pairwise nonisomorphic finitely presented non-MF groups, and continuum many pairwise nonisomorphic finitely generated non-MF groups. The groups $E \times \mathbb{Z}^k$ for $k \geq 0$ are finitely presented, pairwise nonisomorphic, and non-MF, each containing E (Lemma S6.7(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks. For finitely generated examples, choose continuum many pairwise nonisomorphic finitely generated groups N, for instance Neumann's two-generator groups [Ne]. Each product $E \times N$ is non-MF. A fixed countable group X can be isomorphic to $E \times N$ for at most countably many isomorphism types of N, because each such N embeds in X as a finitely generated subgroup and X has only countably many finitely generated subgroups. Hence the products yield $2^{\aleph_0}$ pairwise nonisomorphic finitely generated non-MF groups.

S10. Limitations of the operator-norm method

The argument requires operator-norm control. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm: the estimate $\|\mathrm{Ad}(U) - \mathrm{Ad}(V)\| \leq 2\,\|U - V\|$ has no converse, and there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints are at operator-norm distance 2. Operator-norm control is needed here to ensure that the classes $[\mathrm{Ad}\, U_n(g)]_\omega$ are multiplicative; Hilbert–Schmidt control alone does not provide this.

Adding an identity block leaves operator-norm multiplicative defects and separations unchanged but decreases normalized Hilbert–Schmidt distances. By taking the identity block sufficiently large, all pairwise normalized Hilbert–Schmidt distances can be made smaller than any prescribed positive bound. Thus universal 2-norm triviality of an element does not by itself prove a group non-MF. The central-involution argument passes to an invariant corner and equips it with the trace normalized by its own rank r_n: the 2-norm separation on the corner is measured against r_n rather than d_n, so r_n/d_n may tend to zero without affecting the contradiction. The spectral projection of Theorem S3.7 is normalized on its own range in the same way, and the proof of Theorem S7.1 normalizes by the rank k_n.

Stable finiteness and faithful traces on the coordinate algebras do not suffice for Theorem 3.4. For a finite factor M in its standard representation on $L^2(M, \tau)$, the right action is the commutant of the left action. Hence the von Neumann algebra generated by the two actions is $(M \cup M')'' = Z(M)' = B(L^2(M, \tau))$; in infinite dimension this is not a finite C^*-algebra. Constant coordinates $A_n = C^*_{\mathrm{r}}(E)$ give an explicit counterexample: the left regular representation embeds E exactly and injectively into the norm quotient

$$\prod_n A_n / \bigoplus_n A_n,$$

the element c centralizes the Kazhdan base there, and conjugation by t produces tct^{-1}, whose commutator with the base is the element u with $u^2 = w \neq 1$ of Theorem A. Hence Theorem 3.4 fails when its matrix coordinates are replaced by stably finite unital C^*-algebras carrying faithful tracial states and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Their question concerns Hamming and tracial models, whereas the adjoint spectral estimates here require operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply. The soficity of E is proved separately, from its normal form (Theorem D).

Unitary lifting is not matrix-specific. If u is unitary in $\prod_n A_n / \bigoplus_n A_n$ for unital C^*-algebras A_n, and (x_n) is any bounded lift, then $\|x_n^* x_n - 1\| \to 0$ and $\|x_n x_n^* - 1\| \to 0$. Thus x_n is invertible for all large n, and the polar correction $u_n = x_n (x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \to 0$. This lifting step needs only polar correction in the coordinate algebras, with no real-rank-zero or semiprojectivity hypothesis in the sense of Loring [Lo98]. The matrix-specific inputs in Theorem 3.4 are instead the Hilbert-space conjugation model and finiteness of the norm ultraproduct.

Questions.

(1) Is there a torsion-free finitely presented non-MF group? For torsion-free groups, Theorem S3.3 gives no obstruction because every finite subgroup is trivial, while Theorem S3.7 does apply. What remains is group-theoretic: a torsion-free finitely presented group whose subgroup N_{conj} contains a *nontrivial* normal property-(T) subgroup. Remark S10.1 gives a small-cancellation route over the torsion-free finitely presented Kazhdan group of [FFF], which removed the analogous torsion obstacle for nonsoficity. The Clifford construction of Theorem 5.1 cannot produce such a group: it imposes $c^2 = 1$, so $\hat{c}^2 = 1$, and a torsion-free quotient q then has $q(\hat{c}) = 1$ and $q(N_{\mathrm{conj}}) = 1$.

(2) Does MF imply hyperlinearity? An MF embedding controls operator norm but does not require the induced tracial model to separate nontrivial group elements, so it does not directly produce a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.

(3) Does the obstruction survive when the matrix blocks are replaced by other building blocks? The proof of Theorem 3.4 uses finite dimensionality of $M_{d_n}(\mathbb{C})$ for the conjugation action on $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and for finiteness of the norm ultraproduct; faithful traces alone do not replace these inputs. For which separable C^*-algebras A_n does every homomorphism $H \to \mathcal{U}(\prod_n A_n / \bigoplus_n A_n)$ still map w to the identity?

(4) For which groups do the criteria of Theorems S3.3, S3.7 and S7.1 compute the residual exactly?

(5) What is $\mathrm{Res}_{\mathrm{MF}}(E)$? It properly contains $\{1, w\}$ by Theorem B, and Corollary S7.5, applied to $E/\langle w \rangle$ and its defect $D = D_{\mathrm{coll}}$, computes it from $\mathrm{Res}_{\mathrm{MF}}((E/\langle w \rangle)/D)$. These reductions give no further information about that last group: $w = 1$ in $E/\langle w \rangle$, and the generators of D are trivial in the quotient by D. That quotient may still contain a finite normal subgroup, a normal Kazhdan subgroup, or an orbit configuration to which the preceding criteria apply.

Remark S10.1 (a small-cancellation realization). The affine base of Sections 7–8 can be replaced by the small-cancellation construction of [FFF, §2], which Fournier-Facio pointed out to us. There a universal finitely presented torsion-free group — one containing a copy of every finitely presented torsion-free group — is embedded by small cancellation in a finitely presented torsion-free property-(T) group P. Hence that P contains every finitely presented torsion-free group, in particular a direct product $P_1 \times P_2$ with $P_i \cong P$; loc. cit. states this as $P_1 \times P_2 \times S \leq P$ with S finitely presented simple, and the third factor is not needed here. Choosing an isomorphism $\alpha\colon P \to P_1 \leq P$ and any $a \in P_2 \setminus \{1\}$ gives the input required by Theorem 5.1. The group $E(P, \alpha, a)$ has torsion even though P does not, since the central involution w of Theorem 5.1 is nontrivial; adjoining a generator with $c^2 = 1$ would on its own leave open that c collapses in the presented quotient.

S11. The maximal group C^*-algebra

For a countable group H, the maximal group C^*-algebra $C^*_{\max}(H)$ is the completion of the group ring $\mathbb{C}[H]$ in the norm

$$\|x\|_{\max} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of H; the supremum is finite because $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$. Write u_h for the canonical unitary of $h \in H$.

Proposition S11.1 (the universal property)**.** *The canonical map $h \mapsto u_h$ is injective, and for every unital C^*-algebra B and every homomorphism $\rho\colon H \to \mathcal{U}(B)$, there is a unique unital $*$-homomorphism $C^*_{\max}(H) \to B$ with $u_h \mapsto \rho(h)$ for every $h \in H$.*

Injectivity follows from the left regular representation, and uniqueness follows from density of $\mathbb{C}[H]$.

Proposition S11.2 (a strictly dominated projection gives a proper isometry)**.** *Let A be a unital C^*-algebra containing a projection p and a unitary u with $p < upu^*$. Then A contains an isometry that is not a unitary; consequently A is not stably finite and has no faithful tracial state.*

Proof. Write $q = upu^*$, so q is a projection with $pq = qp = p$, and put $r = pu^*$. Then

$$r^*r = up\,pu^* = upu^* = q, \qquad rr^* = pu^*up = p.$$

Put $s = r + (1-q)$. Since $p(1-q) = 0$ and $(1-q)p = 0$,

$$s^*s = r^*r + (1-q) = q + (1-q) = 1,$$

and since $pu^*(1-q) = pu^* - pu^*upu^* = pu^* - pu^* = 0$ and likewise $(1-q)up = 0$,

$$ss^* = rr^* + (1-q) = p + (1-q) = 1 - (q-p).$$

Thus $s^*s = 1$ but $ss^* = 1-(q-p) \neq 1$, so s is a nonunitary isometry. A unital C^*-algebra containing a nonunitary isometry is not finite, hence not stably finite. If τ were a faithful tracial state then $\tau(1-ss^*) = \tau(1)-\tau(s^*s) = 0$ while $1 - ss^* = q - p$ is a nonzero positive element, contradicting faithfulness. □

Proposition S11.3 (the maximal algebra under proper one-sided conjugation)**.** *Let G be a countable group containing a property-*(T) *subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ is not stably finite, has no faithful tracial state, and is in particular neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the spectral projection at 1 of the Hermitian Kazhdan average $\frac{1}{2|S|}\sum_{s\in S}(u_s + u_s^*)$, which exists by the spectral gap of Section 3. Then $P \le u_t P u_t^*$, and the inequality is strict: in the quasi-regular representation on $\ell^2(G/t\Gamma t^{-1})$ the point mass at the base coset is fixed by $t\Gamma t^{-1}$ but moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections are distinct in this representation. Now apply Proposition S11.2. □

The base subgroup of E and the element t form such a pair, so $C^*_{\max}(E)$ is not stably finite. The reduced algebra behaves differently: its canonical trace is faithful and it is stably finite, despite being non-MF (Theorem C).

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distinct roles of the operator and Hilbert–Schmidt norms. We thank Alon Dogon for pointing out that the interaction between those two norms also occurs in the earlier approach of Bachner–Dogon–Lubotzky [BDL].

AI and computational resource usage

A supercomputer was used to rule out possible approaches but did not contribute to the mathematics used in this paper. GPT-5.6 Sol and Claude Fable 5 were used in Claude Code, Codex, and on the ChatGPT and Claude websites over multiple days, across thousands of user prompts, and Claude Opus 5 also with formalization and operational coding tasks. A custom informal proof tool was used to provide scaffolding to the AIs. The non-MF result was first formulated by GPT-5.6 Sol ("high" thinking effort, ChatGPT web) while working on the question of if all groups are hyperlinear, after the user mentioned the possibility of the existing work applying to MF groups instead of the hyperlinear problem. This instance was given thousands of AI-written pages of informal notes accumulated from multiple days of work on related problems, including an informal proof that MF groups are not contained in sofic groups, in addition to over 80 previous user messages. Other parallel agents without the MF idea (including "Pro" mode) did not discover the key MF result. The exact prompt given to the instance leading to the non-MF proof was:

> "Achieve f{MF of the Clifford Kun–Thom extension }$\widetilde{W}$ in such a way that it archives our goals and ends everything, or, possibly, a different breakthrough, leading to achieving the final result. You got this!"

The preceding suggestion to work on MF groups was:

> "Anyway. Are we closer to are all groups hyperlinear or are all groups MF? Proof or disproof either way. Should we switch to MF?"

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